DEGREE PROJECT IN MECHANICAL ENGINEERING,
SECOND CYCLE, 30 CREDITS
STOCKHOLM, SWEDEN 2021

Simulation of Mechanical Shock with Finite Element Analysis and Estimation of Shock Attenuation

CHENG QIAN

KTH ROYAL INSTITUTE OF TECHNOLOGY
SCHOOL OF ENGINEERING SCIENCES

Simulation of Mechanical Shock with Finite Element Analysis and Estimation of Shock Attenuation

CHENG QIAN

Master thesis in Solid Mechanics
Date: June 30, 2020
Supervisor: Gunnar Tibert
Examiner: Gunnar Tibert
School of Engineering Sciences
Swedish title: Finit elementanalys av mekanisk chock och uppskattning av chockdämpning

Abstract

Space equipment is often subjected to mechanical shocks. Some devices like electronics are sensitive to shocks, and they can easily be damaged. To protect these devices, it is important to study shock penetrations in space structures. The Finite Element Method is an effective tool to simulate response to transient excitations. Unlike Statistical Energy Analysis (SEA) which can ensure sufficient accuracy in high frequency range, Finite Element Analysis (FEA) is limited to low frequency range, but has no spatial restrictions. In this thesis, Tuma's digital filter method and Irvine's recursive digital method were combined to calculate the Shock Response Spectrum (SRS). FEA models based on a given experimental system, i.e., shock table were used to predict the structural response to Haversine shaped forces. The modal transient structural analysis in ANSYS Workbench was used as the solver. Transient analysis based on modal results neglects material and structural non-linearities, so it uses less memory and computation time. FEA and the Bernoulli beam equation were employed to simulate the shock response of cantilever beams and fixed-pinned beams to validate the FEA models. SRS calculated from FEA results were compared with those from the beam equation results. Almost no difference between Bernoulli beam equation results and FEA results for thin beams reveals that the FEA models were validated. The SRS of two beam models calculated from FEA using solid elements were compared with those using beam elements. The results from two different element types are almost consistent with each other. Response at different positions on the shock table were measured to predict the shock attenuation. The attenuation was described in the way of remaining percentage from shock source, and the curve from FEA simulation roughly agreed with the attenuation rule from the ESTEC database.

Key words: shock, SRS, FEA, attenuation.

Sammanfattning

Rymdutrustning utsätts ofta för mekaniska stötar, s.k. mekanisk chock. Vissa enheter, som elektronik, är känsliga för stötar och de kan lätt skadas. För att skydda dessa enheter är det viktigt att studera chockdämpningen i rymdstrukturer. Finita elementmetoden är ett effektivt verktyg för att simulera responsen vid transienta excitationer. Till skillnad från statistisk energianalys (SEA) som kan säkerställa tillräcklig noggrannhet i högfrekvensintervall, är finit elementanalys (FEA) begränsad till lågt frekvensområde, men har inga rumsliga begränsningar. I detta examensarbete kombinerades Tumas digitala filtermetod och Irvines rekursiva digitala metod för att beräkna Chockresponsspektrum (SRS). FEA-modeller baserade på ett givet experimentellt system, d.v.s. chockbord, användes för att förutsäga den strukturella responsen på Haversine-formade krafter. Den modala transienta strukturanalysen genomfördes med ANSYS Workbench. Den transienta analysen baserad på modala resultat försummar icke-linjäriteter i material och struktur för att använda mindre minne och förkorta beräkningstiden. FEA och Bernoulli-balkekvationen användes för att simulera chockresponsen av konsolbalkar och balkar med fast inspänning i ena änden och fri uppläggning i den andra änden, för att validera FEA-modellerna. SRS beräknat från FEA-resultaten jämfördes med resultat från balkekvationen. Ingen avgörande skillnad noteras mellan Bernoullis balkekvationsresultat och FEA-resultat för tunna balkar, vilket validerar FEA-modellerna. SRS för två balkmodeller beräknade från FEA med solida element jämfördes med de som använde balkelement. Resultaten från de två olika elementtyperna överensstämmer mycket bra. Responsen vid olika positioner på chockbordet mättes för att förutsäga chockdämpningen. Chockdämpningen beskrivs som kvarvarande procent av responsen vid chockkällan och kurvan från FEA-simulering överensstämde ganska bra med tumregeln för chockdämpning från ESTEC-databasen.

Nyckelord: chock, SRS, FEA, chockdämpning.

Content

1. Introduction

1.1. Mechanical shock in spacecraft

During the launch and the separation of a rocket, satellites are exposed to different types of loads such as vibrations, thermal expansion, steady state acceleration and shock. [1] Among these loads, deterministic loads can be expressed as functions of time, while random loads can only be predicted statistically. [2] These loads can cause deformation and damage on the structure or the electronic devices in the satellites. Therefore, it is important to deal with different loadings in the design and the testing of spacecrafts.

Mechanical shocks are transient loadings on a structure that result in significant displacement. Shocks can be generated from launch, separation, transportation, and other forms of movement of rockets. [3] Although shocks have very short period of time, they can cause serious damage to the systems as well as the critical components. [4] For example, impact on electronics can bring malfunction due to degradation of electronic components. The mechanisms under impact can get cracks, fractures and fatigue and thus leading to malfunction. [2] Hence, the shock impulse during the separation and launch is an important factor to determine the strength and the reliability of the space structures. [5]

1.2. Literature survey on mechanical shock

Many researchers have studied the shock testing on spacecraft-related products. Most of the studies focused on the shock environment and the measurement instruments. García-Pérez et al. analyzed the in-plane excitation shock test with 6 triaxial and 3 unidirectional accelerometers using the finite element method. They stated advantages and disadvantages of different analysis methods according to the test results. [6] Morais and Vasques developed a pendulum in-plane shock test with a mono-plate supported at 6 points. They created a reliable finite element modeling methodology for simulation and prediction on shock tests. [7] Kunicka conducted the shock dynamic analysis based on simulations on the FE model of the Vega shock test apparatus. She optimized the SRS by parameterizing different factors, applied multiple sensors and accelerometers to determine the accuracy of the SRS and used lump mass to predict and improve the shock responses. [8]

Some researchers investigated the shock environment through the pyroshock experiments. Ding et al. investigated the spatial and temporal variations of the acoustic waves produced by the launch of the Shenzhou 10 spacecraft. [9] Demenko and Birukov experimented with the loading on small spacecraft induced by the triggering of separation system by Lavochkin Association. [5] Mayes and Rohe performed physical vibration simulation with 6 shakers on an industrial structure. With this method they were able to nearly replicate an acoustic environment up to about 3500 Hz. [3]

Studies also focused on enhancing performance of vibration control for electronic devices and transportation facilities. Ahmed and Abdelkefi developed a reduced-order model to simulate the shock response and observed the mechanical resistance performance of single and dual-beam systems to mechanical shocks. [10] Niwa et al. attached both horizontal and vertical rubber springs as vibration isolation system to reduce both the high frequency vibration and the low frequency vibration. [11] George et al. used electrodynamics shaker to generate excitation signal, and applied it on virtual instrument and on LabVIEW to develop a new direction of studying the SRS. [12]

1.3. Attenuation

The study on shock attenuation first started in Europe with the development of the first shock isolator for ARIANE. [13] The occurrence of small spacecrafts in Europe, which were usually interfaced to launchers through discrete interface points, increased the interest in shock attenuation. [13] In an aerospace structure, the shock attenuation mainly depends on path length, orientation between structure elements and joint types. Some organizations such as NASA have proposed a number of shock attenuation rules. Mostly the shock attenuation rules are derived from empirical data. These rules might not match well with every specific case, but they can still be taken as reference. In the cases of point source excitations, the attenuation rules are best represented by the attenuation of ramp and plateau in the SRS, with distance from the source. [14][15][16]

The attenuation rule of thumb and the measurement of attenuation will be discussed in chapter 2.5.

1.4. Background of the project

This study is based on the Miniature Student Satellite project (MIST). [17] In this project a 3U CubeSat was built, which contained top, mid and bottom stacks. The satellite is characterized by 3 units with a total dimension of $300 \times 100 \times 100\ \mathrm{mm}^3$, and each unit has a standard dimension of $100 \times 100 \times 100\ \mathrm{mm}^3$. The MIST CubeSat is shown in Figure 1.1.

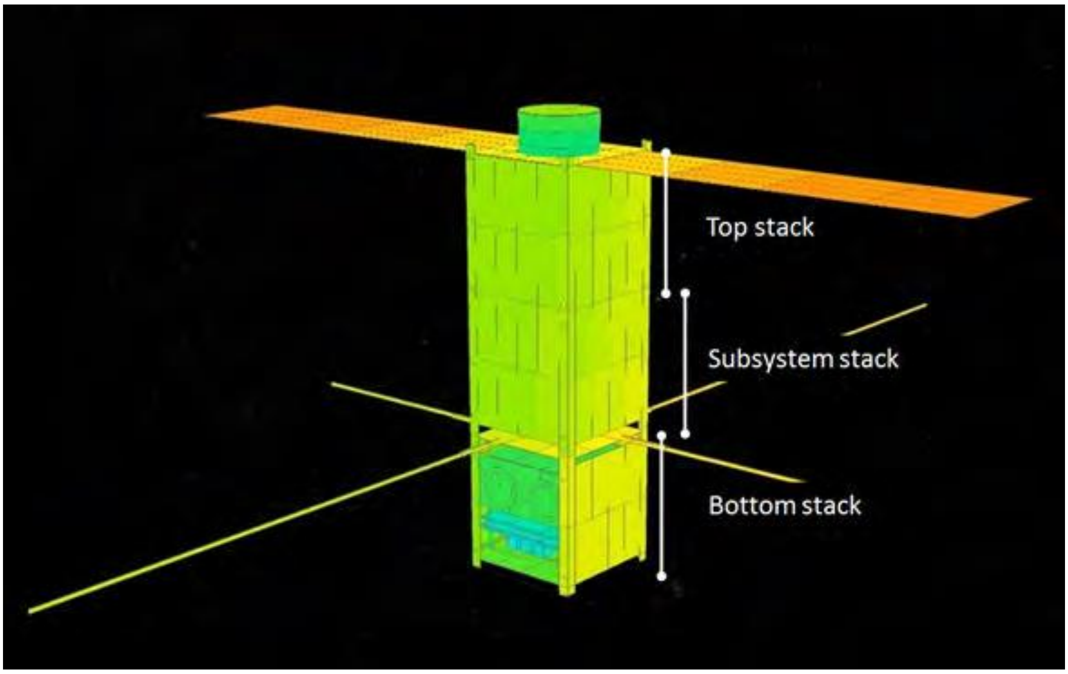


Figure 1.1: Overview of MIST [17]

The subsystems and other payloads are placed in the 3 units. Most of the subsystems are contained in the middle unit of the satellite, so the mid stack is also called subsystem stack. The structure and the subsystems are mostly bought off-the-shelf from ISIS. The subsystems include:

- An onboard computer, which is the brain of the satellite.
- Signal transmitting and receiving devices.
- Deployable sonar panels and power system.
- Devices for connecting experiments and control systems.

The details of the subsystem stack are shown in Figure 1.2.

The CubeSat contains shock-sensitive devices. The shock generated by the launch and separation of the rocket may cause a high risk of damage. Therefore, the aim of this thesis is to analyze the shock response behavior of the aerospace structures and provide suggestions on the structural development and hardware positioning.

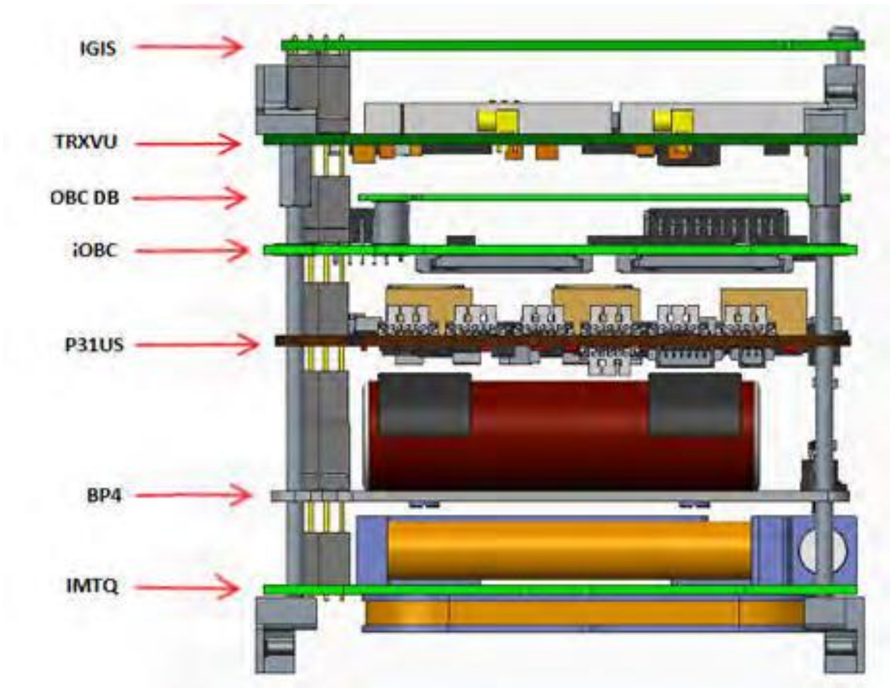
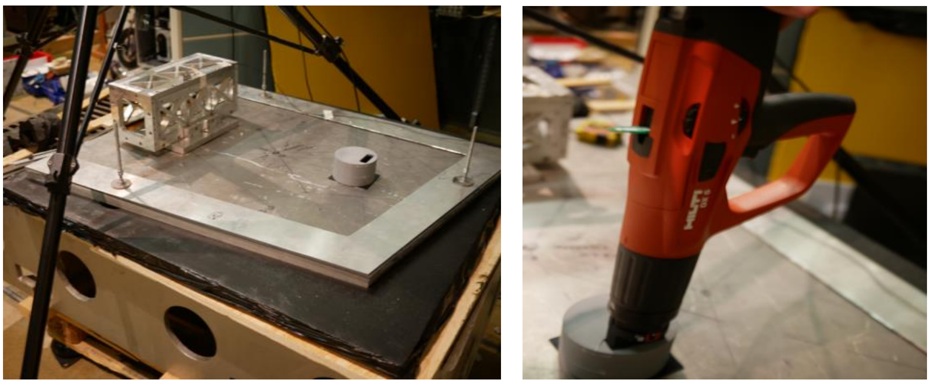


Figure 1.2: MIST "subsystem stack". From the bottom: iMTQ =Magnetic torque coils and magnetometer, BP4=Battery pack, P31us=Power supply, iOBC=Onboard computer, OBC DB=OBC Daughterboard, TRXVU=Transceiver, IGIS=ISIS Generic Interface System. [17]

1.5. Motivation

An experimental system, a shock table was designed for pyroshock test of MIST, which is shown in Figure 1.3. Figure 1.3 (a) shows the shock table and a model of the CubeSat, and Figure 1.3 (b) displays the nailgun that was utilized as excitation. Results from simulations on the shock table were compared with the experimental data to provide evidence for future studies.



(a) (b)

Figure 1.3: Experiment system of shock table for MIST CubeSat

The structure of the shock table was complicated and the shock response of the shock table could hardly be described by theoretical formulas. To better understand the structural response to pyroshocks, it is a good way to start from simple models such as beams and plates. Besides, the validity of FEA models is required to be verified. Therefore, numerical calculation based on the Bernoulli beam equation was used to calculate the beam vibration, and the results were compared with beams built by FEA models.

To know more about how the shock penetrates in space structures, the shock attenuation was studied by testing the responses at different positions on the shock table and expressed by plotting the response level vs the distance from shock source. The results were compared with the attenuation rules from ESTEC criteria.

1.6. Scope: limitations

In this study, the shock table was only simulated by an FEA model because of its complicated structure. Meanwhile, modal transient analysis was chosen as the solver. This solver neglects non-linearities in materials and structures, so it requires less memory. For convenience, the shock table is built with a simplified model which possesses a smaller size and lacks intricate contacts between surfaces. These steps make computation easier, though they lead to lower accuracy in results.

Theoretical calculation was performed on only two beam models. The beams and shock table differ greatly in structures, so the investigation on the SRS of the former provided limited reference on the response of the latter. The purpose of studying the SRS of beams is to verify the FEA models. To acquire more concepts on shock response, studies on other simple geometries can be taken in future works.

The influence of materials and joints on shock attenuation were not investigated in this project. However, these factors are also very important in the study of pyroshocks. Therefore, the conclusion of shock attenuation regulars in this thesis might not be thorough.

2. Theory and Methods

In this section, we tried to analyze the response behavior of the shock table under pyroshock, and to investigate the effects of important factors on the shock response level. A combination of Tuma's and Irvine's algorithms was used to calculate the shock response spectrum (SRS) for engineering purposes. Theoretical calculations of beam response vibrations and finite element analysis will be employed to predict the pyroshock environments that the CubeSat may encounter.

2.1. Prediction of pyroshock environments

In general, 3 ways (including analytical models, direct measurements and extrapolations from previous measurements) could be used to predict the shock response at different positions on a structure. [18]

Various analytical methods have been developed to estimate the response of aerospace structures to the transient loads. For example, the finite element method (FEA) and statistical energy analysis (SEA) are widely used to transfer pyroshock energy into mid- and far-fields. [14] It is believed that FEA models are limited to a low frequency range and simple structural configurations, while SEA models can better predict mid frequency responses. However, SEA uses spatial and spectral averaging, so it is not suitable for acquiring shock response at specific locations or frequencies. [2] [13] [14]

In this thesis, we focused on the prediction of shock responses at different positions on a structure, which means the study is spatial dependent. Therefore, finite element analysis is preferred in this work.

2.1.1. Pyroshock environment

The pyroshock environment exerts much influence on the shock response behavior. The pyroshock environment can be divided into 3 categories: near-field, mid-field and far-field. For most aerospace structures the difference between the categories is the magnitude and the spectral content of the environment. Several factors can affect these properties, such as the strength of the pyroshock device, the distance between the hardware and the excitation source, and the structural details like joints, corners and lumped mass, etc. The environment categories are descried as follows [13] [14]:

a) The near-field environment: this environment is under the direct wave propagation from the source. The peak acceleration of the response can exceed 10 000 G and the spectral content is above 10 000 Hz. Pyroshock-sensitive hardware should not be exposed to this environment.

b) The mid-field environment: this is an environment dominated by a combination of both wave propagation and structural resonance. The peak acceleration differs from 1 000 G to 10 000 G and the substantial spectral content is between 3 000 Hz and 10 000 Hz.

c) The far-field environment: it is dominate by structural resonance which lead to a peak acceleration below 1 000 G and spectral content below 3 000 Hz.

Both the structural design and the hardware positioning influence the pyroshock attenuation significantly. The pyroshock environment category provides an effective guideline on aerospace installation and hardware design.

2.2. Shock response spectrum

The primary accessible data is time history since all physical signals are obtained by recording variations with time. [19] Although the time history experimental data contains all the pertinent information in time domain, it cannot be directly used for engineering purposes. For example, it does not show difference on shock severities and frequency contents. It does not help us understand the effects of shock on the hardware. Thus, digital signal processing tools are needed to deal with shocks.

The influence of shock loads on a structure is usually described by an SRS. An SRS is a plot that presents the relationship between the largest response and the natural frequency when a single degree of freedom system is excited with shock loads. The idea is to apply a base excitation in time domain on a series of single degree of freedom (SDOF) systems and calculate the maximum response of each of them. It is derived by calculating the maximum response of a SDOF system to a transient excitation. For a series of SDOF systems the same input time history is used. The natural frequencies of these SDOF systems are independent. Figure 2.1 shows an SRS model combined with a series of SDOF systems on an excited base. Each SDOF system is defined with a damping ratio, and the value $\xi = 0.05, Q = 10$ (quality factor) is commonly used. A typical time history shock signal and its SRS is shown in Figures 2.2 and 2.3. [14][19]

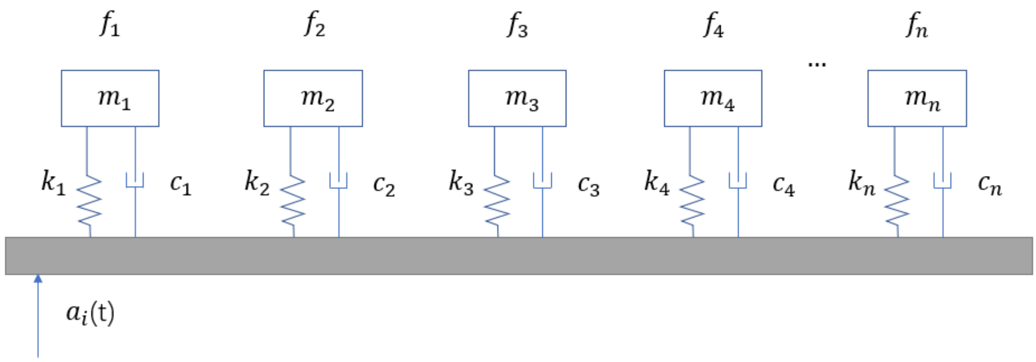


Figure 2.1: The SRS model

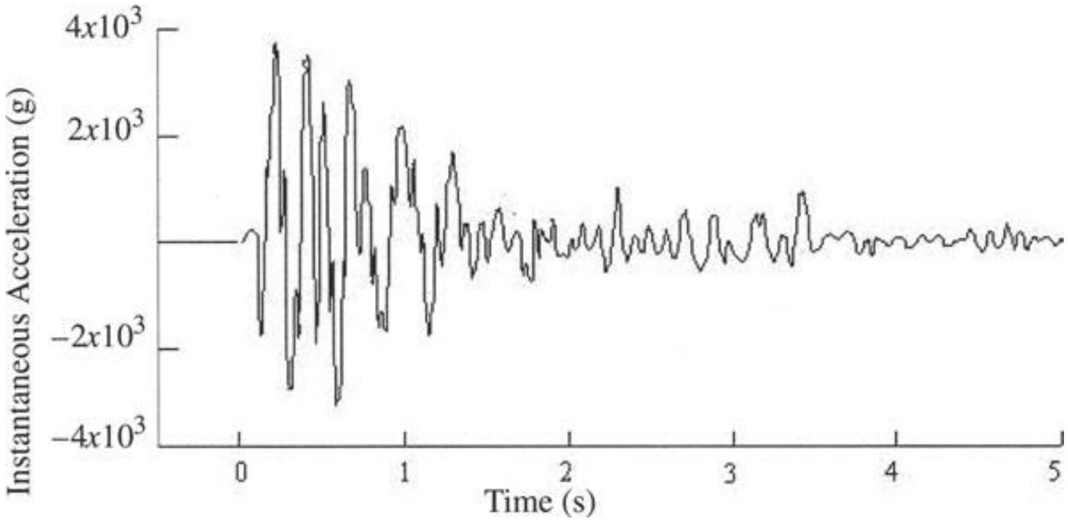


Figure 2.2: Typical pyroshock acceleration time-history [14][19]

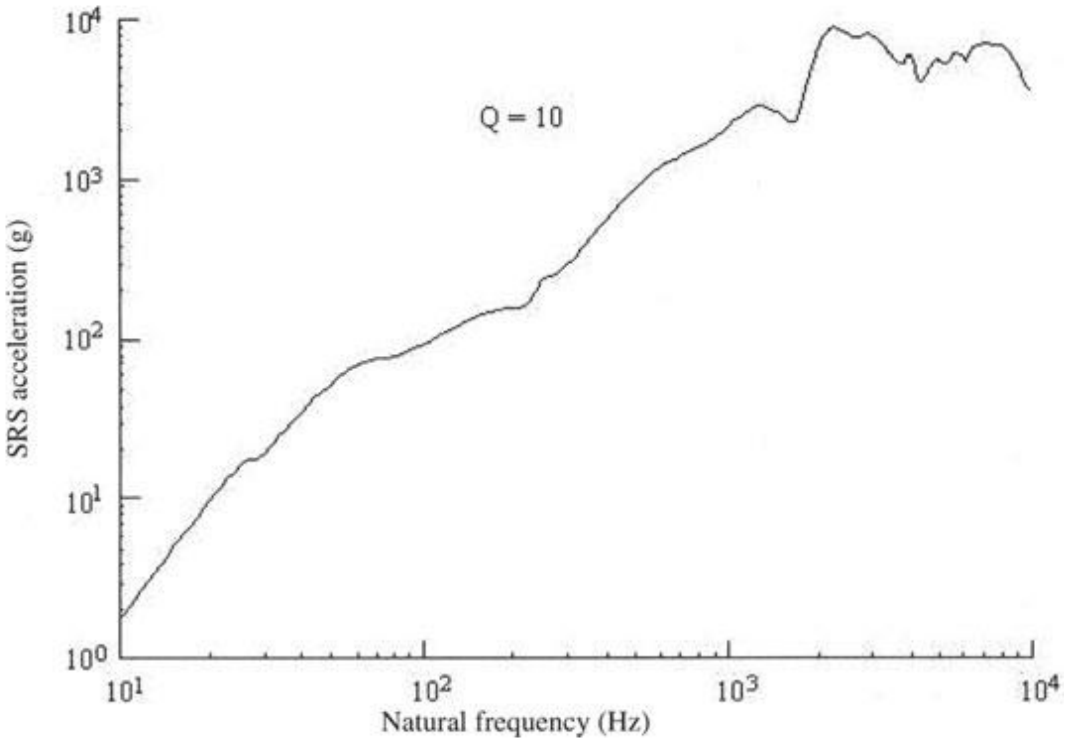


Figure 2.3: Typical pyroshock maximum shock response spectrum (SRS) [14][19]

As mentioned above, a series of natural frequencies are used to calculate the shock response spectrum, and these frequencies are arbitrary. A generally used way to define the frequencies is based on a proportional bandwidth, such as 1/3, 1/6 or 1/12 octave, etc. Take 1/12 octave as an example. When using 1/12 octave frequency band, is means that each successive frequency is $2^{1/12}$ times the pervious one. The SRS resolution is dependent on this bandwidth but not on the signal duration. In the present study, 1/6 octave bandwidth is used for calculating the SRS of different models.

In chapter 2.2.1 the deduction of the SRS for an enforced SDOF system will be presented. In chapter 2.2.2 two SRS algorithms will be introduced.

2.2.1. The SRS for an enforced SDOF system

A theoretical calculation of the SRS for SDOF system is introduced by Wijker. [19] An SDOF system is a basic mechanical vibration system. It normally contains a mass, a spring and a damper. A typical sketch of a damped SDOF system is shown in Figure 2.4. The system has a mass connected to the base by a spring and a damper.

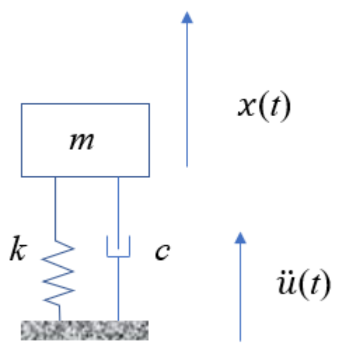


Figure 2.4: Enforced acceleration on a damped SDOF system

Assume that the base is movable, and the base is moving with an acceleration $\ddot{u}(t)$. Then a relative displacement between the mass and the base can be defined as

$$z(t) = x(t) - u(t) \tag{1}$$

The motion of the system can be rewritten as

$$\ddot{z}(t) + 2\xi\omega_n\dot{z}(t) + \omega_n^2 z(t) = -\ddot{u}(t) \tag{2}$$

where $\xi = \frac{c}{2\sqrt{km}}$ is the damping ratio and $\omega_n = \sqrt{\frac{k}{m}}$ is the natural angular frequency of the system. The absolute acceleration can be written as

$$\ddot{x}(t) = \ddot{z}(t) + \ddot{u}(t) = -2\xi\omega_n\dot{z}(t) - \omega_n^2 z(t) \tag{3}$$

Assume that the initial relative displacement and velocity are $z(0) = \dot{z}(0) = 0$, the expression of the relative displacement can be derived as

$$z(t) = -\int_0^t \mathrm{e}^{-(\xi\omega_n\tau)} \frac{\sin\omega_d\tau}{\omega_d} \ddot{u}(t-\tau)\mathrm{d}\tau = -\int_0^t \mathrm{e}^{-(\xi\omega_n(t-\tau))} \frac{\sin\omega_d(t-\tau)}{\omega_d} \ddot{u}(\tau)\mathrm{d}\tau \tag{4}$$

The relative velocity can be derived by differentiation

$$\dot{z}(t) = -\int_0^t \mathrm{e}^{-(\xi\omega_n(t-\tau))} \cos\big(\omega_d(t-\tau)\big)\, \ddot{u}(\tau)\mathrm{d}\tau - \xi\omega_n z(t) \tag{5}$$

Introduce the equations (4) and (5) into (3), and the absolute acceleration becomes

$$\ddot{x}(t) = 2\xi\omega_n \int_0^t \mathrm{e}^{-(\xi\omega_n(t-\tau))} \cos\big(\omega_d(t-\tau)\big)\, \ddot{u}(\tau)\mathrm{d}\tau + \omega_n(2\xi^2 - 1)z(t) \tag{6}$$

For each natural frequency, the maximum absolute acceleration $\ddot{x}(t)$ can be calculated by inserting $\omega_n = 2\pi f_n$ and be plotted with respect to the frequency f_n. The plot is the shock response spectrum which responds the excitation acceleration $\ddot{u}(t)$.

The shock response is defined as the maximum acceleration for each frequency

$$S = |\ddot{x}(t)| \tag{7}$$

The shock response for a damped SDOF system is

$$S = \left\{2\xi\omega_n \int_0^t \mathrm{e}^{-(\xi\omega_n(t-\tau))} \cos\big(\omega_d(t-\tau)\big)\,\ddot{u}(\tau)\mathrm{d}\tau + \omega_n(2\xi^2-1)z(t)\right\}_{\max} \tag{8}$$

2.2.2. The SRS calculation algorithm

2.2.2.1. Digital filter

This SRS algorithm was developed by Tuma and Koci. [20] As mentioned in chapter 2.2.1, the SRS of a SDOF system is derived from the equation of motion of the system, which is stated in equation (2). Substitute the base acceleration with a base displacement, the system is displayed in the following way:

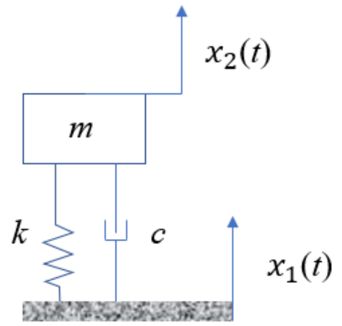


Figure 2.5: Enforced displacement on a damped SDOF system

The equation of motion can be written as:

$$m\ddot{x}_2 = -c(\dot{x}_2 - \dot{x}_1) - k(x_2 - x_1) \tag{9}$$

where x_1 is the base displacement and x_2 is the response displacement. The corresponding velocities and accelerations as time functions are v_1, v_2, a_1, a_2 respectively. The Laplace transfer function of the system which relates the base and the response displacement can be written as

$$G(s) = \frac{X_2(s)}{X_1(s)} = \frac{cs+k}{ms^2+cs+k} \tag{10}$$

The Laplace transform of the velocities and the accelerations can be described in the form of Laplace displacements:

$$V_1(s)=sX_1(s),\ \ V_2(s)=sX_2(s),$$
$$A_1(s)=sV_1(s),\ \ A_2(s)=sV_2(s)\ . \tag{11}$$

Where $\mathrm{L}\{x_1(t)\} = X_1(s), \mathrm{L}\{x_2(t)\} = X_2(s)$ are the Laplace transform of the displacements.

Then the Laplace transfer function of the velocity and the acceleration is:

$$G(s) = \frac{V_2(s)}{V_1(s)} = \frac{A_2(s)}{A_1(s)} = \frac{cs+k}{ms^2+cs+k} \tag{12}$$

Introduce the natural frequency f_n, the amplification factor Q and the damping ratio ξ, and describe them with SDOF system components,

$$f_n = \frac{1}{2\pi}\sqrt{\frac{k}{m}}, Q = \frac{\sqrt{km}}{c}, \xi = \frac{1}{2Q} = \frac{c}{2\sqrt{km}}. \tag{13}$$

Replace the system components in equation (12) with the parameters in (13), the Laplace transfer function of the acceleration can be written as

$$\frac{A_2}{A_1} = \frac{\frac{\omega_n s}{Q}+\omega_n^2}{s^2+\frac{\omega_n s}{Q}+\omega_n^2} = \frac{R_1}{s-s_1} + \frac{R_1^*}{s-s_1^*} \tag{14}$$

Where $\omega_n = \frac{f_n}{2\pi}$ is the circular frequency, R_1 is a residue, s_1 is a pole, R_1^* and s_1^* are the corresponding complex conjugate quantities.

It is practicable to calculate the response acceleration as a function of time, if the amplification factor Q, natural frequency f_n and the base acceleration a_1 as a function of time are known. However, in most cases the base acceleration is sampled with a certain sampling time interval. The Laplace transfer function is then required to be transformed into a z-plane to adapt to a discrete system (digital filter).

Herein a frequency correction is added to avoid frequency distortion:

$$G^*(z) = G(s)|_{s=\frac{2\pi f_p}{\tan(\pi\frac{f_p}{f_S})}\frac{1-z^{-1}}{1+z^{-1}}} \tag{15}$$

Where f_S is the sampling frequency and f_p is the prewarping frequency. The prewarping frequency can be substituted by the resonance frequency.

This discrete approximation of a continuous transfer function is based on a method called Ramp Invariant Method designed by Ahlin, Magnevall and Josefsson. [21] In this method a s-plane transfer function

$$G_1(s) = \frac{1}{s+a} \tag{16}$$

Which has the same form as equation (14), is approximated by a z-plane transfer function

$$G_1(z) = \frac{\beta_{10}+\beta_{11}z^{-1}}{1+\alpha_{11}z^{-1}} \tag{17}$$

Where in equation (16) a is a pole and in equation (17) parameters β_{10}, β_{11} and α_{11} are coefficients of the approximating difference equation of the first order. While the equation (14) shows that the system is a second order system which can be expressed by the decomposition of two parallel first order systems. The discrete approximation of the SDOF system of the second order filter can be expressed by the z-plane transfer function

$$G^*(z) = \frac{A_2(z)}{A_{1(z)}} = \frac{\beta_{20}+\beta_{21}z^{-1}+\beta_{22}z^{-2}}{1+\alpha_{21}z^{-1}+\alpha_{22}z^{-2}} \tag{18}$$

The filter parameters are given by the ISO 18431-4 Standard as following

$$\beta_{20} = 1 - \frac{\exp(-A)\sin(B)}{B} \tag{19}$$

$$\beta_{21} = 2\exp(-A)\left[\frac{\sin(B)}{B} - \cos(B)\right] \tag{20}$$

$$\beta_{22} = \exp(-2A) - \frac{\exp(-A)\sin(B)}{B} \tag{21}$$

$$\alpha_{21} = -2\exp(-A)\cos(B) \tag{22}$$

$$\alpha_{22} = \exp(-2A) \tag{23}$$

where

$$A = \frac{\omega_n T_S}{2Q}\ , \quad B = \omega_n T_S\sqrt{1-\frac{1}{4Q^2}}\,. \tag{24}$$

2.2.2.2. Recursive digital filter

This method was developed by Irvine. [22] In this method a different way (bilinear transform) was used to deal with discrete systems. The link between s-plane and z-plane is:

$$s = \frac{2}{T_s}\frac{(z-1)}{z+1} \tag{25}$$

The Laplace transfer function for the response is the same as equation (10), so the Laplace transfer function for the input is

$$G_i(s) = \frac{s^2+\frac{\omega_n s}{Q}+\omega_n^2}{\frac{\omega_n s}{Q}+\omega_n^2} \tag{26}$$

Replace the variable s with z, the z-plane transfer function is then written as

$$G_i(z) = \frac{\left[\frac{2}{T_S}\frac{(z-1)}{z+1}\right]^2+\frac{\omega_n\left[\frac{2}{T_S}\frac{(z-1)}{z+1}\right]}{Q}+\omega_n^2}{\frac{\omega_n\left[\frac{2}{T_S}\frac{(z-1)}{z+1}\right]}{Q}+\omega_n^2} \tag{27}$$

The form of the digital filter can also be written as

$$G_i(z) = \frac{c_0z^2+c_1z+c_2}{z^2+a_1z+a_2} \tag{28}$$

Solve the filter coefficients by equalizing (27) and (28), the coefficients are

$$a_1 = \frac{2C}{4\xi + C} \tag{29}$$

$$a_2 = \frac{-4\xi + C}{4\xi + C} \tag{30}$$

$$c_0 = \frac{4 + 4\xi C + C^2}{C(4\xi + C)} \tag{31}$$

$$c_1 = \frac{2(-4 + C^2)}{C(4\xi + C)} \tag{32}$$

$$c_2 = \frac{4 - 4\xi C + C^2}{C(4\xi + C)} \tag{33}$$

Where

$$C = T_s \omega_n \tag{34}$$

The digital recursive filtering relationship is

$$\begin{aligned} \ddot{x}_{1(i)} &= -a_1 \ddot{x}_{1(i-1)} - a_2 \ddot{x}_{1(i-2)} \\ &+ c_0 \ddot{x}_{2(i)} + c_1 \ddot{x}_{2(i-1)} + c_2 \ddot{x}_{2(i-2)} \end{aligned} \tag{35}$$

Both Tuma's and Irvine's methods used a digital filter to describe a discrete system. There is some difference between them: Irvine used a bilinear transform to get the discrete transfer function, while Tuma added a frequency correction based on that. In Irvine's methods, a recursive relationship was introduced. In the following studies the SRS will be calculated with the combination of the two methods to help improve the accuracy of the algorithm. The z-transform of this new algorithm had a frequency correction based on Tuma's method, and the recursive relationship in Irvine's method was introduced.

2.3. Theoretical calculation of vibration response

Before testing the shock response of the shock table, it is important to start with simple geometries to understand which factors that will influence the SRS. Calculating the vibration response of the beams with equations helps us better understand the relationship between the dynamic characteristics and the vibration response of a structure, and it helps verify the results from finite element simulations. In this section a theoretical method for calculating the beam response will be introduced. The modal calculation of the beams is based on the theory of vibration on uniform beams of bending which will be described in chapter 2.3.1. The response of the beams derived by applying principle of virtual work will be introduced in chapter 2.3.2. For convenience, we referred to the analysis method as the 'beam equation analysis'.

2.3.1. General solution of mode shape factor

In this section, only the bending modes are considered when calculating the vibration of the beams theoretically. Assume that there is no force applied on the beam, and the damping is neglected. According to D'Alembert's principle [23] the governing equation of the beam vibration is

$$-EI\frac{\partial^4 y}{\partial x^4} = m\frac{\partial^2 y}{\partial t^2} \tag{36}$$

where E is the Young's modulus, I is the second moment of area about the bending axis, m is the mass per unit length.

Since it is assumed that there is no applied force nor damping, the vertical displacement can be written as

$$y = \bar{y}\sin\omega t \tag{37}$$

and the second derivative is

$$\frac{\partial^2 y}{\partial t^2} = -\omega^2\bar{y}\sin\omega t \tag{38}$$

where $\bar{y}$ is a mode shape factor that is based on the variable x. The maximum loading corresponds to $\sin\omega t = 1$. Insert it into equation (38), then $\frac{\partial^2 y}{\partial t^2} = -\omega^2 Y$. The equation of motion can be written as

$$-EI\frac{\partial^4 y}{\partial x^4} = m(-\omega^2\bar{y}) \tag{39}$$

Equation (39) can be transformed into

$$\frac{\partial^4 y}{\partial x^4} = \frac{m\omega^2\bar{y}}{EI} = \beta^4\bar{y} \tag{40}$$

where

$$\beta^2 = \frac{\omega}{\sqrt{\frac{EI}{m}}} \tag{41}$$

The general solution of the mode shape factor is

$$\bar{y} = A\cos\beta x + B\sin\beta x + C\cosh\beta x + D\sinh\beta x \tag{42}$$

The constants in the general solution can be derived according to the boundary conditions of different cases.

Theoretically the deductions of the response of both cantilever beam and fixed-pinned beam are the same, but they differ in the mode shape and the natural frequency. Both beam models have the property with the mode shape parameters as below

$$A = -C = -1,\ B = -D. \tag{43}$$

So the mode shape factor can be written as

$$\bar{y} = -(\cos\beta x - \cosh\beta x) + D(\sin\beta x - \sinh\beta x) \tag{44}$$

In the next chapter the mode shape will be used to get the equation of motion of the beams. To deduct the shock response of beams, a method based on the principle of virtual work was used.

2.3.2. Derivation of equation of motion

One way to get the equation of motion is using the principle of virtual work. [24] The displacement of the beam as a function of x and t is defined in the following way

$$y(x,t) = \sum \phi_i(t) Y_i(x) \tag{45}$$

Where $\phi_i(t)$ are modal displacements that can be determined using principle of virtual work, and the terms $Y_i(x)$ are normalized eigenvectors as functions of x. The subscript i denotes the ith mode to be calculated.

The normalized eigenvectors can be written as

$$Y_i = \sqrt{\frac{1}{\rho AL}}\bar{y}_i \tag{46}$$

where ρ is mass per volume, L is the beam length and A is the cross-section area of the beam.

Replace the mode shape factors $\bar{y}_i$ with equation (42), equation (46) can be written as

$$Y_i = \sqrt{\frac{1}{\rho AL}}\left((\cosh(\beta_i x) - \cos(\beta_i x)) - D_i(\sinh(\beta_i x) - \sin(\beta_i x))\right) \tag{47}$$

The virtual transverse displacements in terms of mode shapes are defined as

$$\delta y_i = \delta\phi_i Y_i \tag{48}$$

Insert the expression of the eigenvectors into equation (48) the virtual transverse displacements become

$$\delta y_i = \delta\phi_i\sqrt{\frac{1}{\rho AL}}((\cosh(\beta_i x) - \cos(\beta_i x)) - D_i(\sinh(\beta_i x) + \sin(\beta_i x))) \tag{49}$$

The work done by inertial force with virtual displacement is

$$\delta W_I = \int_0^L (-m dx)\ddot{y}\,\delta y_i \tag{50}$$

$$\delta W_I = -m\delta\phi_i\int_0^L \ddot{y}\,Y_i(x)\mathrm{d}x \tag{51}$$

$$\delta W_I = -m\delta\phi_i\int_0^L \left\{\sum_{j=1}^{\infty}\ddot{\phi}_J(t)Y_j(x)\right\}Y_i(x)\mathrm{d}x \tag{52}$$

$$\delta W_I = -m\delta\phi_i\sum_{j=1}^{\infty}\ddot{\phi}_J(t)\int_0^L Y_j(x)\,Y_i(x)\mathrm{d}x \tag{53}$$

The orthogonality of the normal mode shapes is

$$\int_0^L Y_j(x)\,Y_i(x)\mathrm{d}x = 0, i \neq j \tag{54}$$

$$m\int_0^L Y_j(x)\,Y_i(x)\mathrm{d}x = 1, i = j \tag{55}$$

The inertial virtual work is

$$\delta W_{Ii} = -\ddot{\phi}_i\,\delta\phi_i \tag{56}$$

The strain energy U due to the elastic forces is

$$U = \int_0^L \frac{EI}{2}(y'')^2\,\mathrm{d}x \tag{57}$$

Replace the displacement y with eigenvectors Y and simplify,

$$U = \frac{EI}{2}\sum_{i=1}^{\infty}\ddot{\phi}_i\sum_{j=1}^{\infty}\ddot{\phi}_j\int_0^L Y_j''(x)\,Y_i''(x)\mathrm{d}x \tag{58}$$

The orthogonality relationships are

$$\int_0^L Y_j''(x)\,Y_i''(x)\mathrm{d}x = 0, i \neq j \tag{59}$$

$$\int_0^L Y_j''(x)\,Y_i''(x)\mathrm{d}x = \beta_i^4, i = j \tag{60}$$

The strain energy can be written as

$$U = \frac{EI}{2}\sum_{i=1}^{\infty}\beta_i^4\phi_i^2 \tag{61}$$

The virtual work of elastic forces is

$$\delta W_{Ei} = -\frac{\delta U}{\delta\phi_i}\delta\phi_i = -EI\beta_i^4\phi_i\delta\phi_i \tag{62}$$

The virtual work of applied forces is

$$\delta W_{Fi} = P(t)\delta y_i(x,t) = P(t)Y_i(x)\delta\phi_i \tag{63}$$

The total virtual work is

$$P(t)Y_i(x)\delta\phi_i - \ddot{\phi}_i\,\delta\phi_i - EI\beta_i^4\phi_i\delta\phi_i = 0 \tag{64}$$

$$\ddot{\phi}_i\,\delta\phi_i + EI\beta_i^4\phi_i\delta\phi_i = P(t)Y_i(x)\delta\phi_i \tag{65}$$

Eliminate $\delta\phi_i$ and insert natural frequency

$$\ddot{\phi}_i + \omega_{ni}^2\phi_i = P(t)Y_i(x) \tag{66}$$

The equation of motion with modal damping is

$$\ddot{\phi}_i + 2\xi_i\omega_{ni}\dot{\phi}_i + \omega_{ni}^2\phi_i = P(t)Y_i(x) \tag{67}$$

Equation (66) indicates the equation of motion at the *i*th mode, and the modal displacements will be solved by inserting initial conditions.

The next step for calculating the response of the beams is to insert the specified mode shape factors and the natural angular frequencies. The calculation of the response to Haversine excitation in time domain of each beam model will be presented later.

2.3.3. Solution of displacement

The modal equation of motion is shown in equation (66). Insert the expression of the excitation force and solve the equation, the modal displacement can be derived. Multiply the modal displacement with mode shape factor, the displacement can be derived.

Assume that

$$P(t) = \begin{cases} \hat{P}\sin(\alpha t)\,, t \le T \\ 0, t > T \end{cases} \tag{68}$$

Where $\alpha = \frac{\pi}{T}$ and T is the duration of the Haversine signal.

The equation of motion at x is

$$\ddot{\phi_i} + 2\xi_i\omega_{ni}\dot{\phi_i} + \omega_{ni}^2\phi_i = \hat{P}Y_i(x)\sin(\alpha t) \tag{69}$$

Let

$$F_i = \hat{P}Y_i(x) \tag{70}$$

We get

$$\ddot{\phi_i} + 2\xi_i\omega_{ni}\dot{\phi_i} + \omega_{ni}^2\phi_i = F_i(x)\sin(\alpha t) \tag{71}$$

Assume that the initial conditions are zero, the modal displacement during the half-sine pulse is

$$\phi_i(t) = \frac{F_i(x)}{\left(\alpha^2-\omega_{ni}^2\right)^2+(2\xi_i\alpha\omega_{ni})^2}\{-2\xi_i\omega_{ni}\alpha\cos(\alpha t) - \left(\alpha^2 - \omega_{ni}^2\right)\sin(\alpha t)\} +$$
$$\frac{1}{\omega_{di}}\left\{\frac{\alpha F_i(x)}{\left(\alpha^2-\omega_{ni}^2\right)^2+(2\xi_i\alpha\omega_{ni})^2}\right\}\{\mathrm{e}^{-\xi_i\omega_{ni}t}\}\{2\xi_i\omega_{ni}\omega_{di}\cos(\omega_{di}t)\} +$$
$$\frac{1}{\omega_{di}}\left\{\frac{\alpha F_i(x)}{\left(\alpha^2-\omega_{ni}^2\right)^2+(2\xi_i\alpha\omega_{ni})^2}\right\}\{\mathrm{e}^{-\xi_i\omega_{ni}t}\}\{\left[\alpha^2 + \omega_{ni}^2(-1+2\xi^2)\right]\sin(\omega_{di}t)\} \tag{72}$$

The modal displacement after the pulse is

$$\phi_i(\tau) = \mathrm{e}^{-\xi_i\omega_{ni}\tau}\left\{\phi_i(T)\cos(\omega_{di}\tau) + \left[\frac{\dot{\phi}_i(T)+(\xi_i\omega_{ni})\phi_i(T)}{\omega_{di}}\right]\sin(\omega_{di}\tau)\right\} \tag{73}$$

$$\tau = t - T$$

The displacement is expressed as

$$y(x,t) = \textstyle\sum_{i=1}^{\infty}\phi_i(t)Y_i(x) \tag{74}$$

The response displacement as a function of time is derived, and the velocity and the acceleration at x can be derived by differential

$$v(x,t) = \frac{\delta y(x,t)}{\delta t}$$
$$a(x,t) = \frac{\delta v(x,t)}{\delta t}. \tag{75}$$

It should be emphasized that the x in equation (69) and in equation (72) denote different positions. The former denotes where the force is applied, and the latter means where the vibration features are measured.

For different cases of beams, the mode shapes can be obtained by introducing the corresponding boundary conditions. The specific solution for cantilever beams and fixed-pinned beams are now discussed.

2.3.3.1. Vibration response of cantilever beams

The cantilever beams are fixed at one end and free at the other end. The structure of a cantilever beam is shown in Figure 2.6. Assume that the length of a cantilever beam is L.

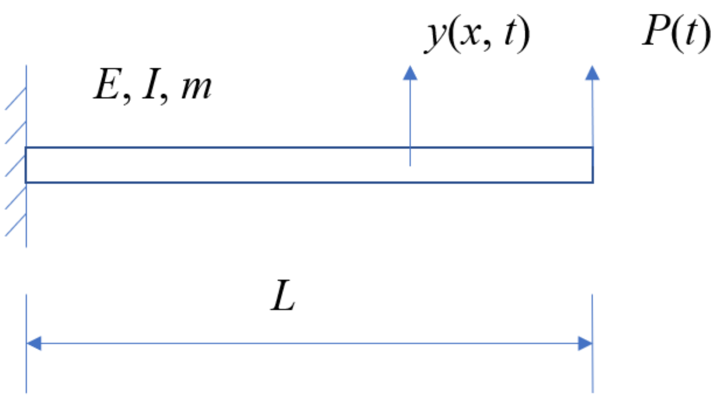


Figure 2.6: Schematic illustration of cantilever beam

The mode shape factor, the eigenvector and the natural angular frequencies are derived at $x = L$, i.e.: the variable x in eigenvector $Y_i(x)$ is replaced with the constant L.

Assumed the beam is fixed at x=0 and free at x=L. The boundary condition is therefore:

$$At\ x = 0, y = 0, \frac{dy}{dx} = 0 \tag{76}$$

$$At\ x = L, \frac{d^2y}{dx^2} = 0, \frac{d^3y}{dx^3} = 0 \tag{77}$$

Solve equation (42) by inserting the boundary conditions, the constants can be found in Table 2.1:

Table 2.1: Characteristic equation, roots and mode shape parameters of cantilever beams [23]

Characteristic equation and roots $\beta_i L$	A_i	B_i	C_i	D_i
$\cos\beta_i L \cdot \cosh\beta_i L = -1$	-1	$\frac{\cos(\beta_i L)+\cosh(\beta_i L)}{\sin(\beta_i L)+\sinh(\beta_i L)}$	1	$-\frac{\cos(\beta_i L)+\cosh(\beta_i L)}{\sin(\beta_i L)+\sinh(\beta_i L)}$
$\beta_i L = \frac{(2i-1)\pi}{2}$				

The natural angular frequencies are

$$\omega_{ni} = \beta_i^2 \sqrt{\frac{EI}{m}} \tag{78}$$

$$\omega_{di} = \omega_{ni}\sqrt{1-\xi^2} \tag{79}$$

For each mode, the mode shape parameters are constant since the roots $\beta_i L$ are known. Insert the parameters and the angular frequencies into equations (47), (71) and (72), the displacement can be obtained.

It should be noted that in this part both the excitation force and the measurement are defined at the free end of the beam. The force, displacement, velocity and acceleration should then be written as special cases at $x = L$, as shown below:

$$F_i = \hat{P} Y_i(L) \tag{80}$$

$$y(L,t) = \sum_{i=1}^{\infty} \phi_i(t) Y_i(L) \tag{81}$$

$$\begin{aligned} v(L,t) &= \frac{\delta y(L,t)}{\delta t} \\ a(L,t) &= \frac{\delta v(L,t)}{\delta t} \end{aligned} . \tag{82}$$

2.3.3.2. Vibration response of fixed-pinned beams

The fixed-pinned beams are defined as beams with a fixed constraint at one end and a pinned constraint on the other. The structure of the fixed-pinned beam is shown in Figure 2.7.

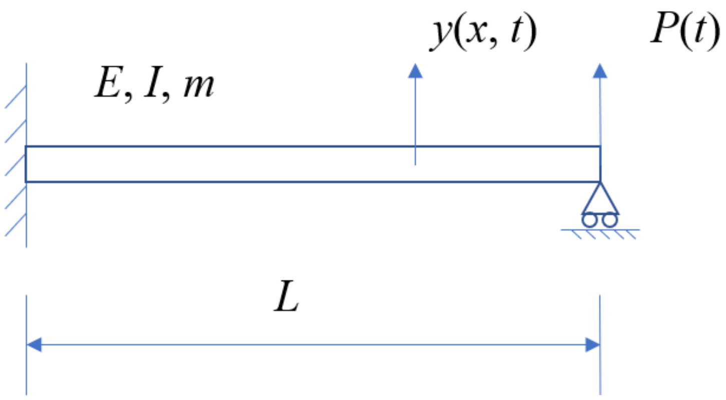


Figure 2.7: Schematic illustration of fixed-pinned beam

Assume that the length of a beam is L, and the beam is fixed on one end at x=0 and pinned at x=L, which means that the boundary condition of the beam can be written as below:

$$\text{At } x = 0, y = 0, \frac{dy}{dx} = 0 \tag{83}$$

$$\text{At } x = L, y = 0 \tag{84}$$

The constants in the general solution of mode shape are derived as shown in Table 2.2:

Table 2.2: Characteristic equation, roots, and mode shape parameters of fixed-pinned beams [23]

Characteristic equation and roots $\beta_i L$	A	B	C	D
$\tan\beta_i L - \tanh\beta_i L = -1$ $\beta_i L = \frac{(4i+1)\pi}{4}$	1	$-\frac{\cos(\beta_i L) - \cosh(\beta_i L)}{\sin(\beta_i L) - \sinh(\beta_i L)}$	-1	$\frac{\cos(\beta_i L) - \cosh(\beta_i L)}{\sin(\beta_i L) - \sinh(\beta_i L)}$

The natural angular frequencies are

$$\omega_{ni} = \beta_i^2 \sqrt{\frac{EI}{m}} \tag{85}$$

$$\omega_{di} = \omega_{ni}\sqrt{1-\xi^2} \tag{86}$$

The excitation is applied at a certain position on the beam, and the response features are measured at a position x along the beam. The mode shapes and modal displacements can then be written as functions of x. The vibration features can be obtained by introducing the mode shape parameters and angular frequencies into equations (73) and (74).

2.4. Finite element analysis on beams and simplified shock table

The finite element analysis (FEA) is an efficient tool which is widely used to evaluate a shock environment in an aerospace structure. The FEA procedures can simulate the vibration response of a structure with relatively high accuracy up to 50th or more modes, if there are sufficient elements.

Before promoting the simulation, appropriate analysis systems should be selected. ANSYS workbench is chosen as the tool in this study. It is a widely used finite element tool which provides strong computational ability and convenient operation. Modal transient structural analysis in workbench can simulate short term structural events and output time-dependent response data. In this analysis system the transient dynamic behaviors of the structures are investigated based on modal results with linear iteration. Therefore, it takes less time but fulfills the requirement of the tasks.

In this section, finite element analysis was taken on beams first. The simulation results of beam response were compared with those from beam equation to verify the FEA models. The influence of length, width and thickness of beams on the SRS was investigated. After that, the simulation was done on the shock table. Two sets of dimensions were used for different purposes. The simulation on the first set was to have a general view on the SRS of the shock table, and that on the second set was to obtain the shock attenuation regular of the shock table.

The geometry parameters and material properties are tabulated in section 3.2. The definition of geometries of different models, the analysis settings, boundary conditions, loading conditions and the element types applied are discussed below.

2.4.1. Geometry: beams and simplified shock table

The geometries of the beams and the shock table will be described. The important steps of building geometries will be specified. Since many sets of sizes are used to define the geometries, some related parameters will not be presented in this part. Instead, they will be tabulated in chapter 3.

2.4.1.1. Geometry of beams

The cantilever beam and the fixed-pinned beam shared the same geometry. They were applied with different boundary conditions and loading conditions which were discussed in section 2.1.3. Two different types of bodies were built to apply different element types.

The first type is the 3-dimensional solid body. The beam is characterized by a constant rectangular cross section along the longitudinal direction. The geometry is built by extruding the cross section and the extent depth is the length of the beam. The geometry of the beam is shown in Figure 2.8 and 2.9.

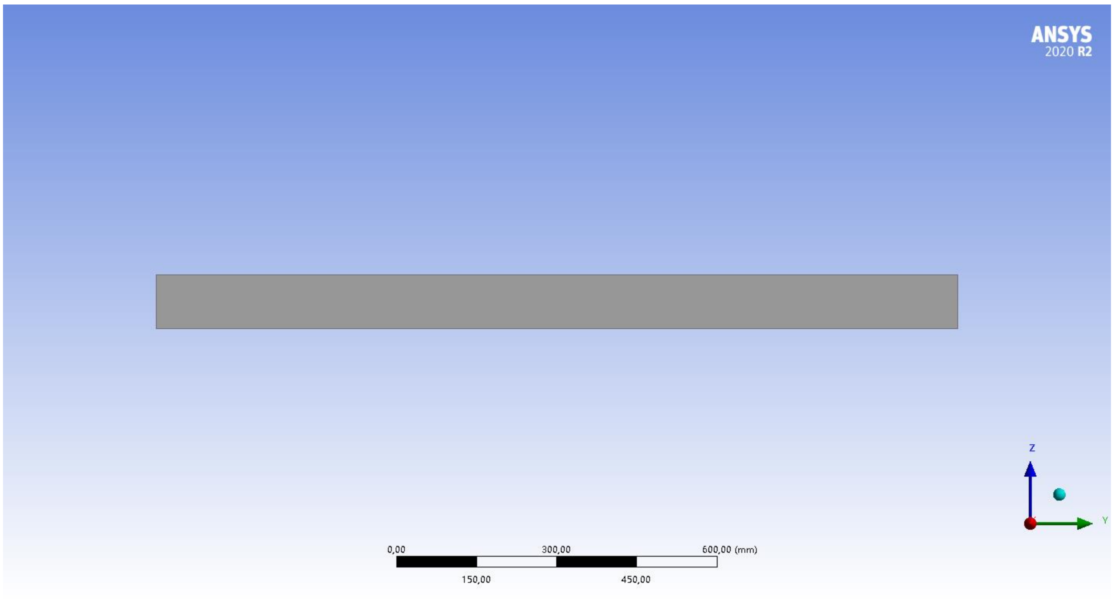


Figure 2.8: Main view of the beam built with solid body

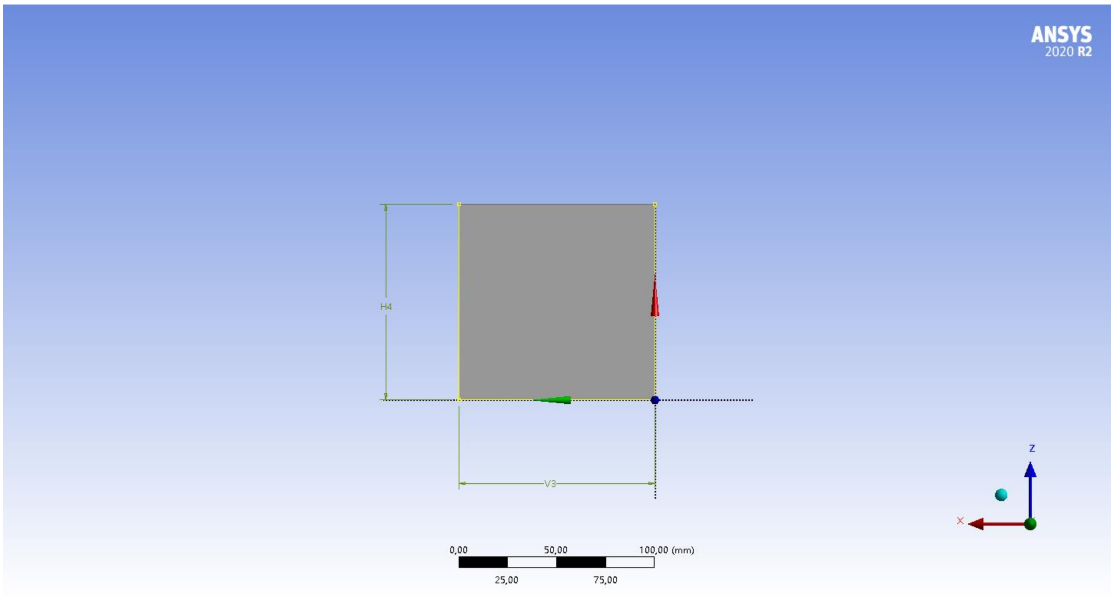


Figure 2.9: Cross section of the beam built with solid body

A series of load points are created on the top surface of the beam for applying loading conditions and acceleration measurements. These points are defined in an array based on the guide edges. These guidelines can determine the start, the end and the array position. The load points on the solid beam are shown in Figure 2.10.

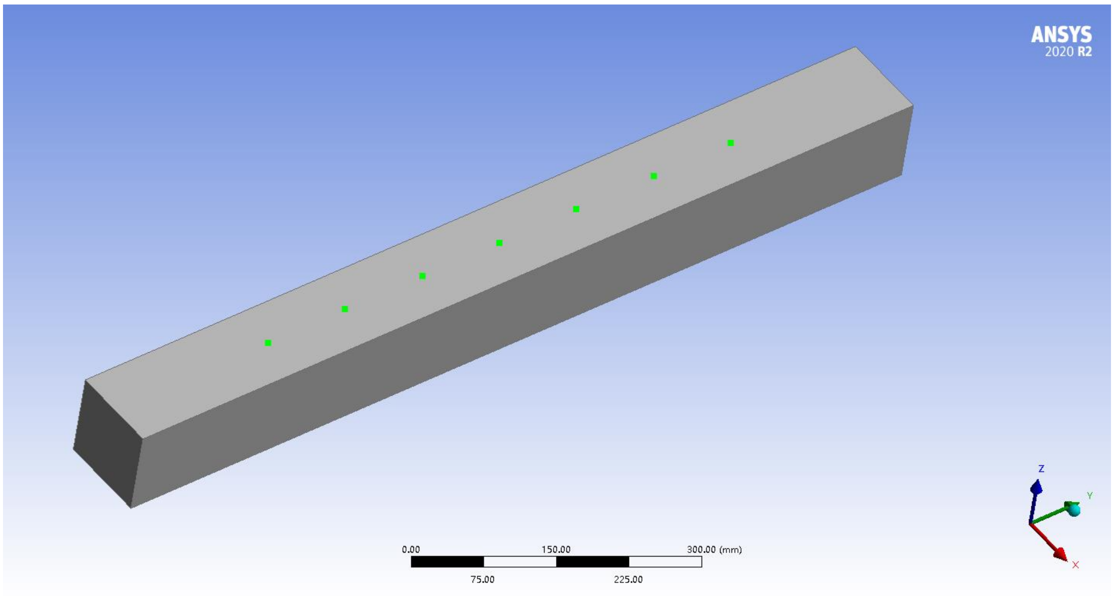


Figure 2.10: Load points on solid beam

The other type is the line body. A straight line is built to define the longitudinal direction. The cross section is defined as a rectangle vertical to the line. The line body and the cross section are shown in Figures 2.11 and 2.12.

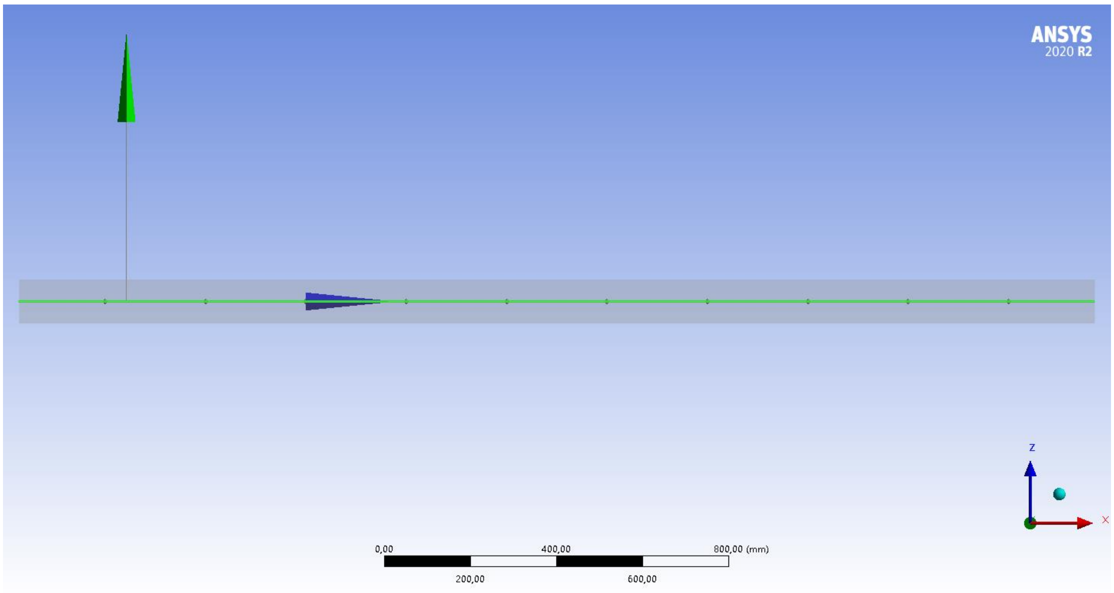


Figure 2.11: Main view of the beam defined with line body

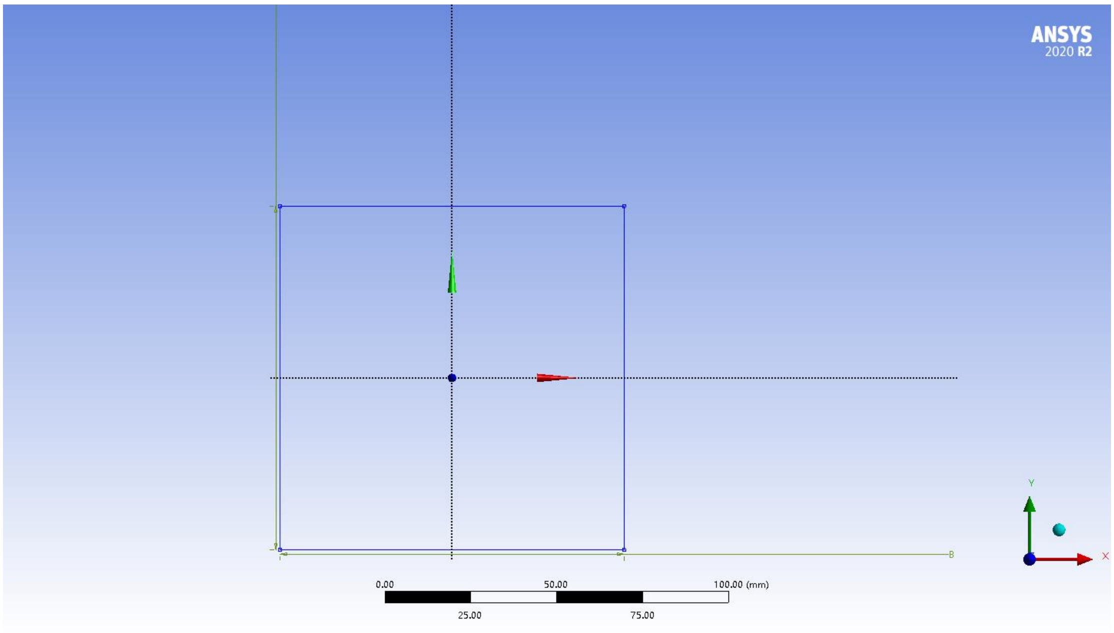


Figure 2.12: Cross section of the beam built with line body

Load points are also created on the line body for the same purpose. The point array is defined by two guide edges that determine the start and the end of the array. The load points on the line body are shown in Figure 2.13.

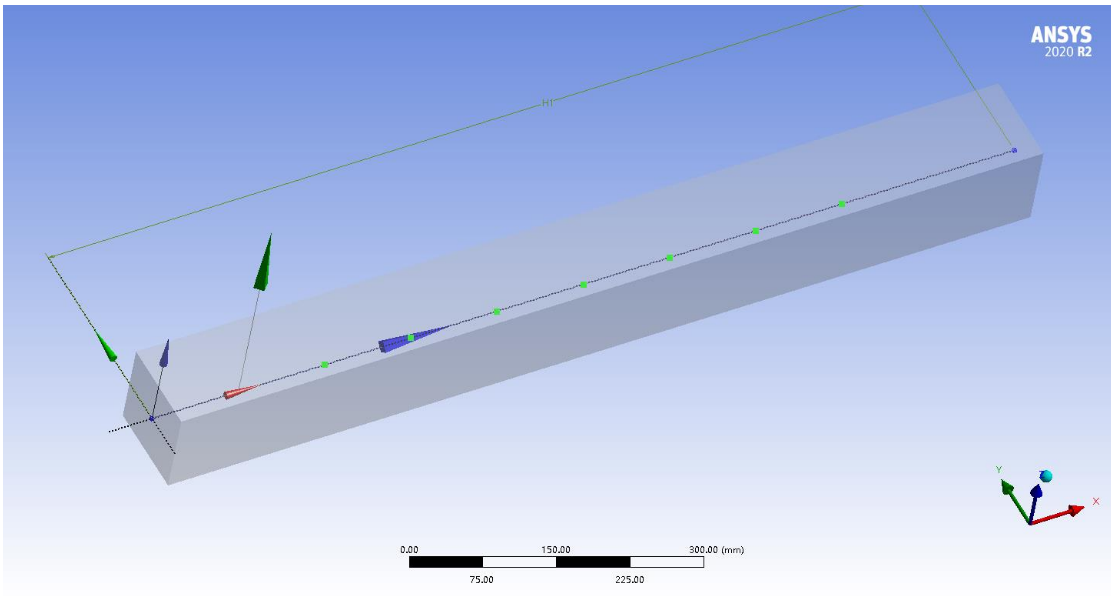


Figure 2.13: Load points on line body

Since different body types were used to build the geometry, different element types were applied.

2.4.1.2. Simplified shock table

The shock table is characterized by a plate-like structure, which is created by extruding a sketch of a rectangle. On the bottom surface of the table, 3 cylinders are extruded from circles as supports. On the top of the shock table a small rectangular plate is created for measuring the response signals, and a cylinder is built to add loading conditions. The main view of the shock table is shown in Figure 2.14.

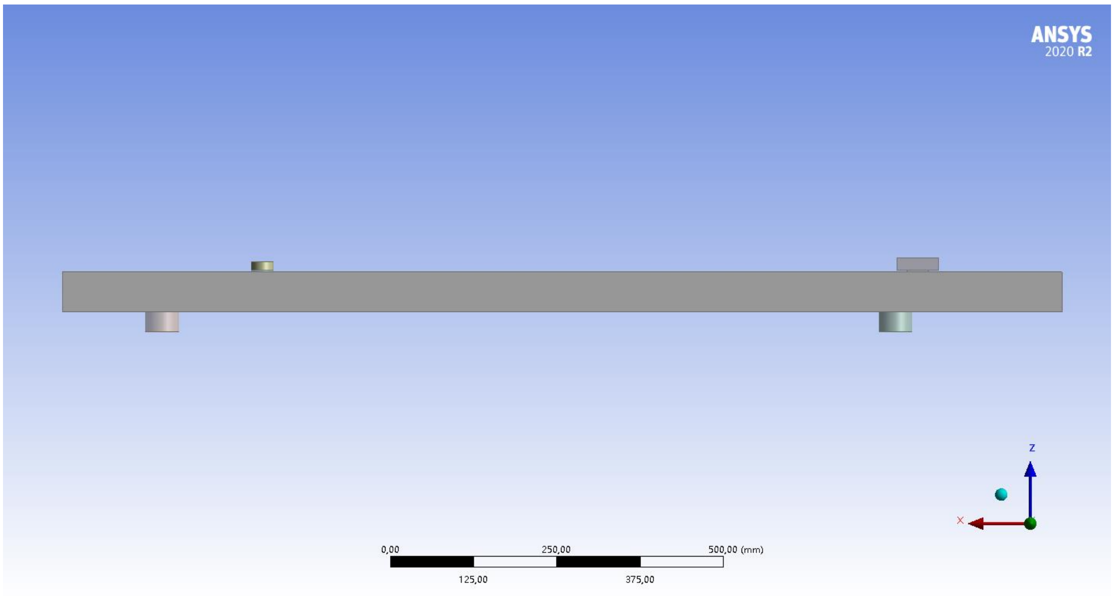
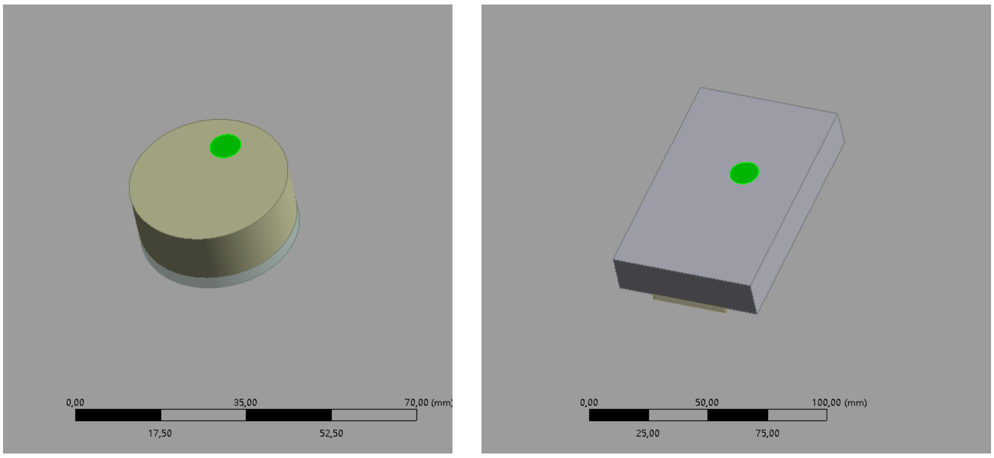


Figure 2.14: the geometry of simplified shock table

At the bottom of the plate and the cylinder, thin damping materials are added to contact with the shock table. On both measurement plate and excitation cylinder, imprint faces are built for applying force and measurement probe. The faces are defined by two sketches of small circles on the top of solid bodies and extruded as imprint faces. The imprint surfaces and the damping materials are shown in Figure 2.15. For simplification, the damping material under the cylinder and the plate will be respectively numbered as 1 and 2 in section 3.2.1 (Figures 3.7 and 3.8).



(a) (b)

Figure 2.15: the imprint face on the excitation

2.4.2. Mesh setting

When analyzing the shock response of the beams, two different types of elements (solid elements and beam elements) are used to study their effect on transient response simulation.

2.4.2.1. Beam built by line body

As mentioned above, the beam geometries are built with different types of solid models, so different element types should be applied to them. The beam modeled by line body is built with beam elements. A beam element has 2 nodes, and there are 6 DOFs on each node. The edge sizing is determined by the number of divisions, and the line is divided into 40 beam elements, as shown in Figure 2.16.

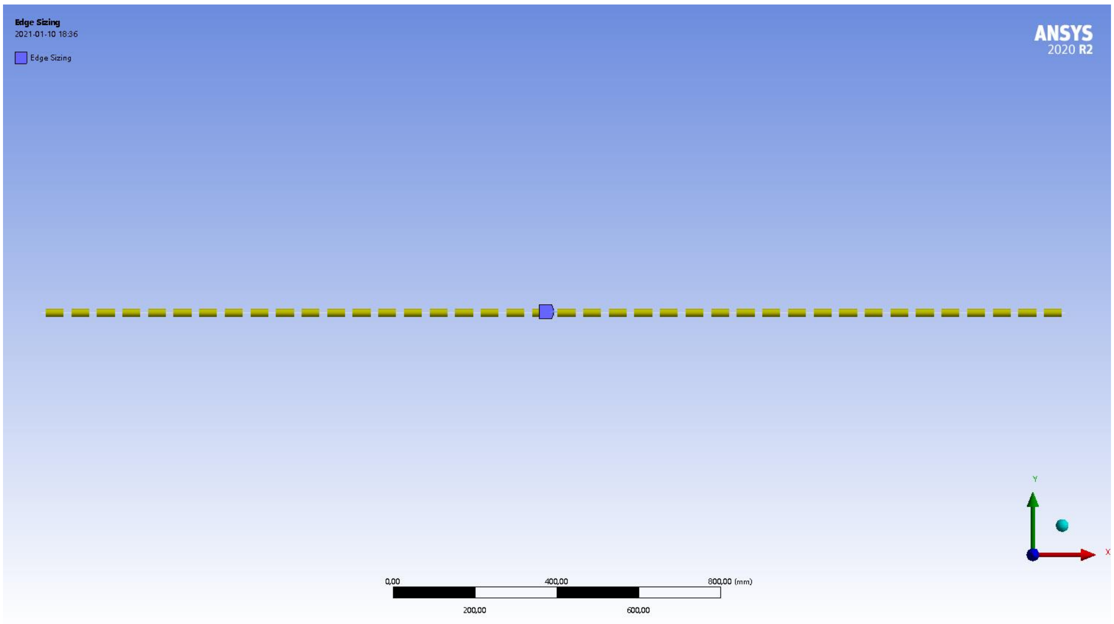


Figure 2.16: beam applied with beam elements

2.4.2.2. Beam built by solid body

The beam was built by solid body using solid elements. The element type is hex dominant mesh with all quad faces. Each element has 8 nodes and each node has 3 DOFs. The element size is set as 30 mm on the cross section, and the mesh is swept along the length. The mesh of the solid beam is shown in Figure 2.17.

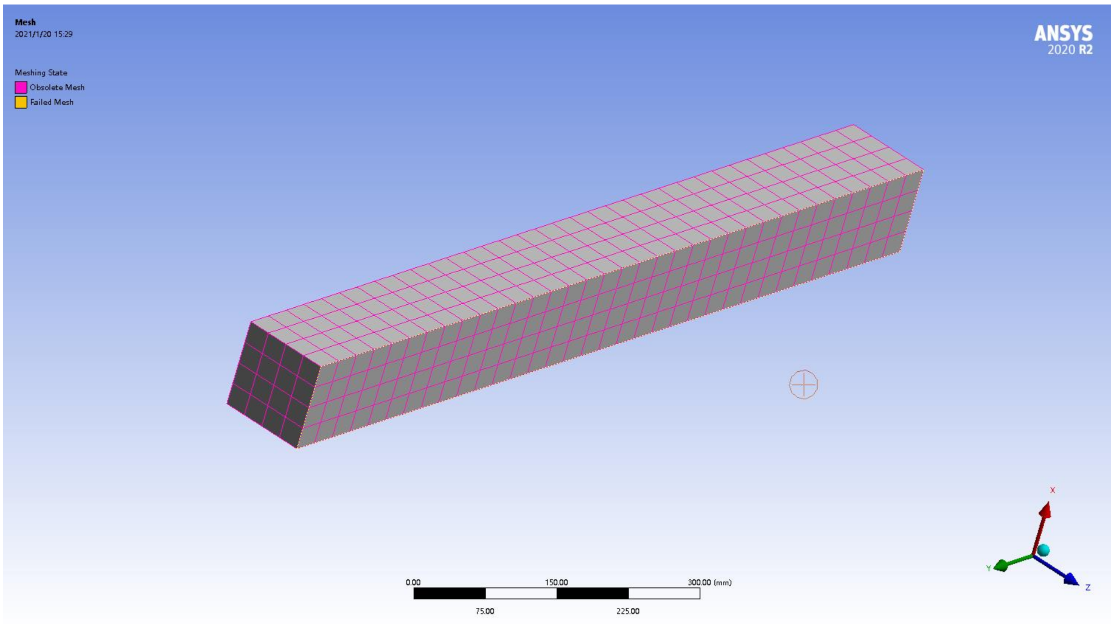


Figure 2.17: beam applied with solid elements

The comparison between the SRS curves from two different element types will be presented in section 3.1.

2.4.2.3. Simplified shock table

In the previous studies on shock beams, it was proven that solid elements in the finite element modal transient analysis is valid (which was discussed in sections 4.1 and 4.2). The conclusion of validating solid elements can be found in section 3.2. In this section, the definition of meshing on different parts of the shock table structure is presented.

Figure 2.18 shows the meshed model of the shock table. Figures 2.19 and 2.20 show the details of mesh on supports, excitation cylinder and measurement plate.

The meshing method applied on the whole geometry is hex dominant, each element has 8 nodes, and each node has 3 DOFs. The table is divided into 3 layers along *z*-direction. The element size on the top surface is 80 mm, and the mesh is swept to the bottom surface.

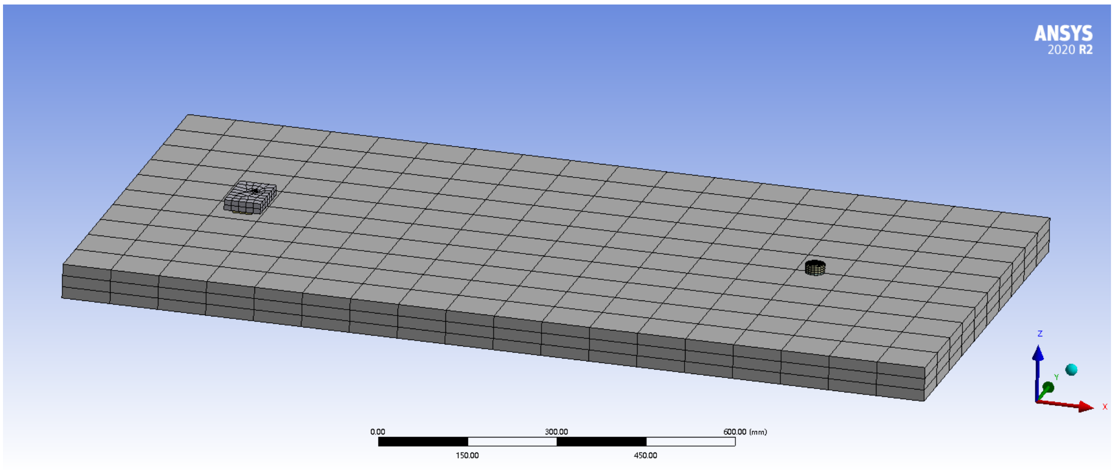


Figure 2.18: Mesh on the shock table

As shown in Figures 2.19 and 2.20, all three supports are meshed with four layers of hexahedron elements, the cylinder and the plate are meshed with three layers and two layers, respectively. Each damping material has one layer of elements.

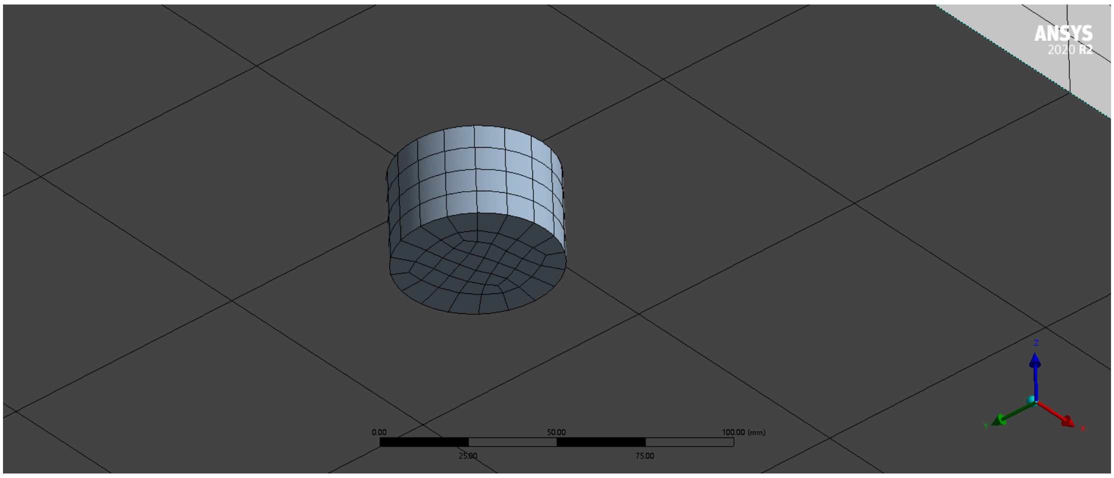
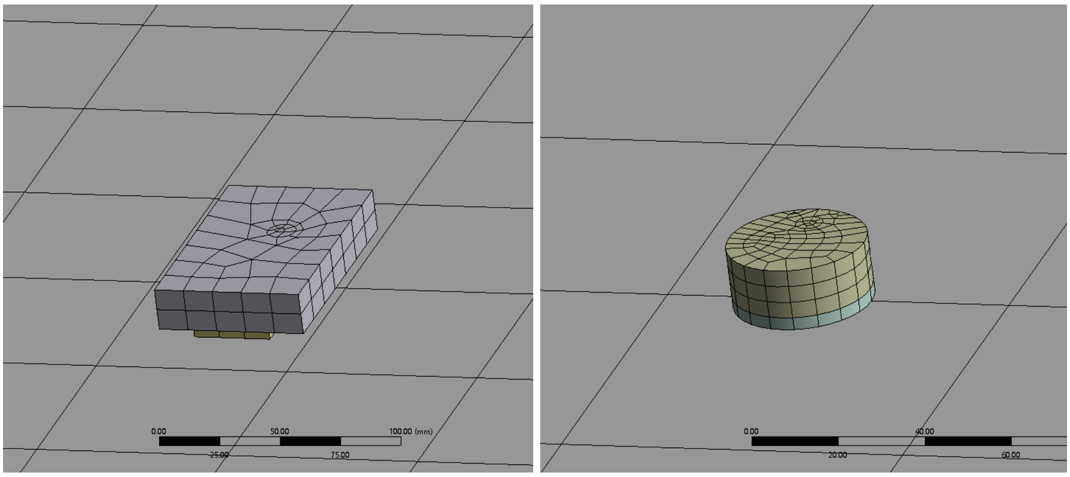


Figure 2.19: Mesh of a support



(a) (b)

Figure 2.20: Mesh of measurement plate (left) and excitation cylinder (right)

2.4.3. Analysis settings (general parameters)

The analysis setting of transient structural analysis will be discussed here. In this study, the same analysis setting will be used for all the FEA simulations.

In modal transient structural analysis, the modal results are pre-conditions of transient simulation. Based on the modal results, the transient simulation removed material and structural non-linearities, so less computation time and memory are required.

The modal result is independent on the solver type (damped or undamped). An undamped solver was used for modal calculation so that the damping ratio could be defined for transient analysis. 50 modes are solved so that it can meet the required frequency range of mid- and far-field of pyroshock environment. In transient analysis, the total sampling time is 0.1 s, the time step is 0.0001 s. The damping ratio is defined as 0.05 for all the modes.

2.4.4. Boundary conditions and loading conditions

This chapter will introduce how the boundary conditions and loading conditions are defined and how they are applied on different geometries. The transient forces applied on three different models are the same, and the force is shown below.

The excitation force of the pyroshock is defined by a Haversine impulse lasting from 0 s to 0.002 s. The maximum magnitude of the impulse is 35 kN, as shown in Figure 2.21. The impulse in the frequency domain transformed by FFT is shown in Figure 2.22. Figure 2.23 shows the SRS of the excitation force.

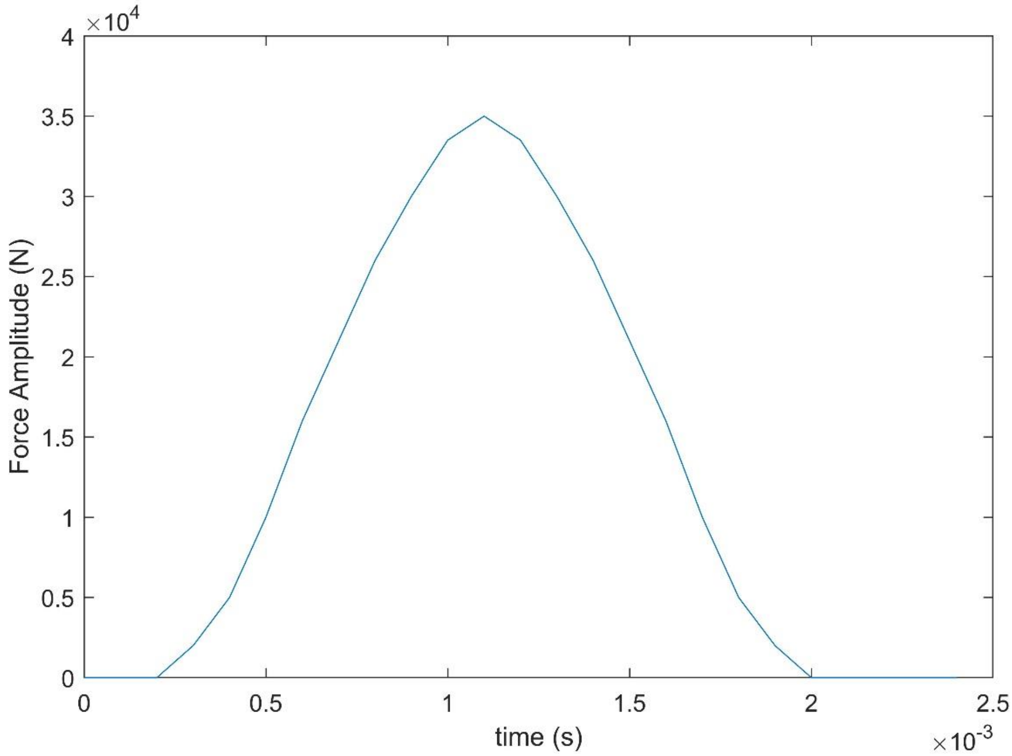


Figure 2.21: Haversine force in time-domain

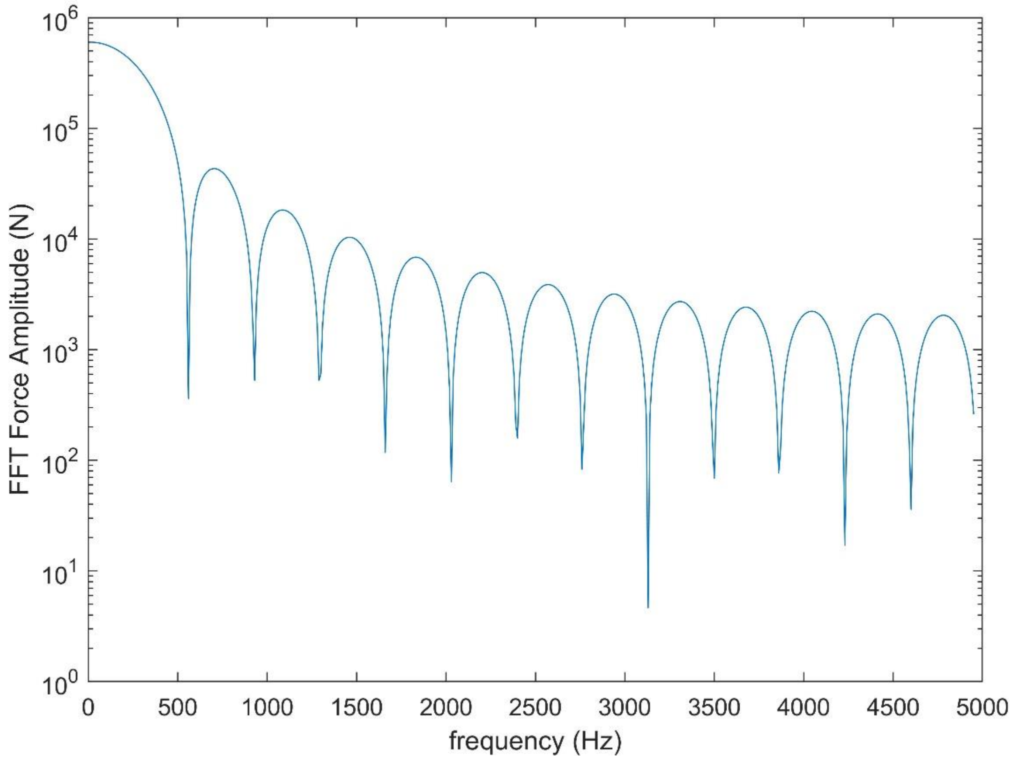


Figure 2.22: FFT of the Haversine force

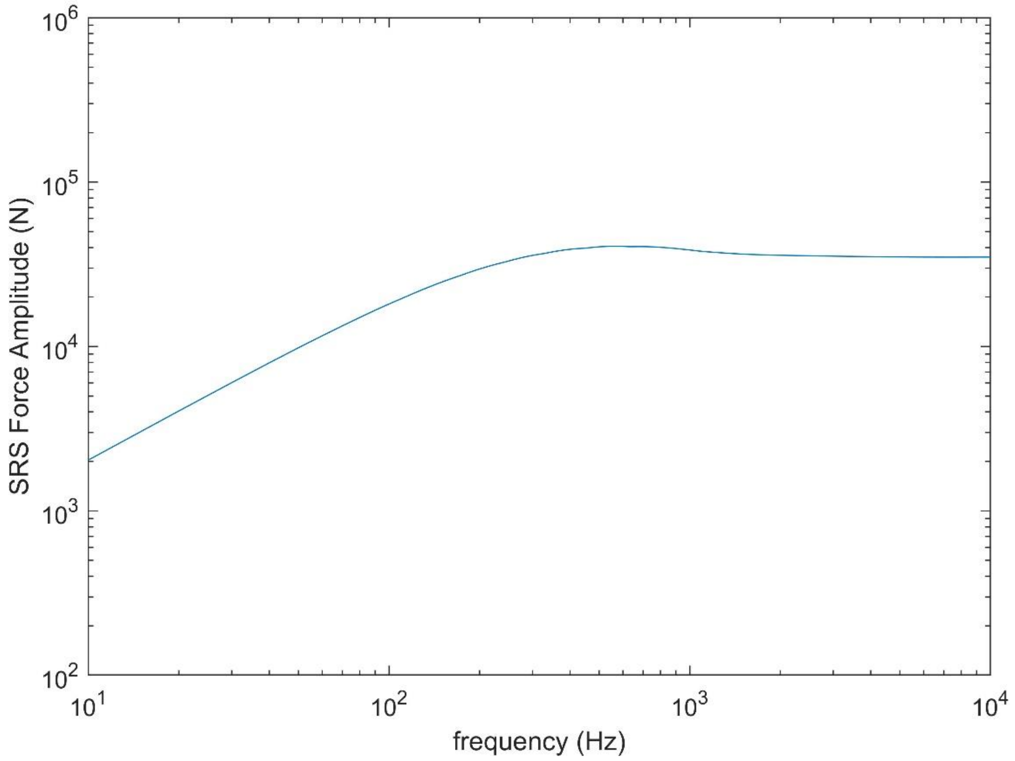


Figure 2.23: The SRS of the Haversine force

2.4.4.1. BC and LC of the cantilever beam

A cantilever beam is defined as a beam with a fixed end and a free end. In this analysis the definition is also based on the element type.

For the beam modeled with beam elements, a fixed support is added on one end of the line body as a point support. The end is restricted in 6 displacement directions. A point force is added at the free end of the beam.

For the beam applied with solid elements, the surface at an end of the beam is fixed supported, with 3 restricted displacement directions. The force is added at the load point in the middle the upper edge of the free end, pointing downwards along z-direction. The definition of the load point was described in section 2.4.1.1.

2.4.4.2. BC and LC of the fixed-pinned beam

A fixed-pinned beam has a fixed end and a pin-supported end. In this part the beam applied with solid elements has a fixed surface on one end, and the surface on the other end is restricted in transversal (x) and vertical (z) directions.

The beam modeled with beam elements is fixed at the point of one end, and simply supported at the point on the other end.

The excitation force is applied at the point $x = 200$ mm, and the acceleration is measured at the point $x = 800$ mm. The direction of the force is reverse of z-direction.

2.4.4.3. BC and LC of the simplified shock table

One of the cylinder supports is fixed at the bottom surface with 3 restricted displacements, and two of them have the simply supported bottom surfaces with 3 constrained translational displacements.

The force is added on the imprint face on top of the cylinder, and the acceleration is measured on the other imprint face on the top of the plate.

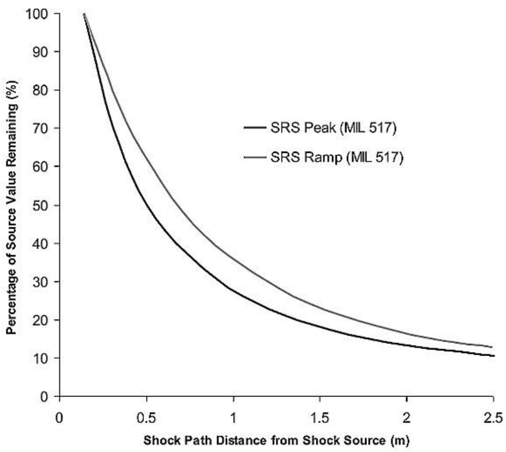
2.5. Attenuation

The pyroshocks have the special characteristics of short duration and high frequency features. Because of these, the influence of the pyroshock will attenuate along the path by the structure. The pyroshock attenuation can be determined by several factors including shock magnitude, joints and structural features. [18]

The attenuation in aerospace structures with distance from the shock source and with joints will be discussed in this chapter. In NASA-HDBK-7005, [18] the attenuation caused by the distance was reflected by the remaining percentage of the response value from the shock source. Response values at two frequencies (spectrum peak and spectrum ramp) are chosen as reference for the comparison. The peak response in an SRS is believed to appear at the knee frequency, so it is the best reference to compare the SRS peak. [25] There is no specified explanation of spectrum ramp found in the handbook, therefore the SRS ramp is defined as a certain frequency in the ramp of the curve. In this thesis, the 20^{th} frequency, which appears around the middle of the ramp, is chosen as the SRS ramp value.

In this part the attenuation rules of thumb will be introduced. These rules of thumb are derived and summarized from empirical data. The attenuation rules from MIL-810-F and ESTEC are illustrated in Figure 2.24 [13]

MIL-810F attenuation rule (method 517)	ESTEC attenuation rule for point source
- Attenuation of SRS peak and SRS ramp with distance from the source - 40% attenuation per joint up to 3 joints (no attenuation in the ramp portion)	- Inspired by MIL method and consolidated with ESTEC shock database - Slightly more conservative than the MIL method - Plateau attenuation: 40% attenuation per joint (1) up to 3 joints - Ramp attenuation: 40% attenuation per joint (1), but counting 1 joint less than for plateau (2) - Percentage of source value remaining to be capped to 100%, for distance lower than 30 cm and 15 cm for SRS Ramp and SRS peak respectively



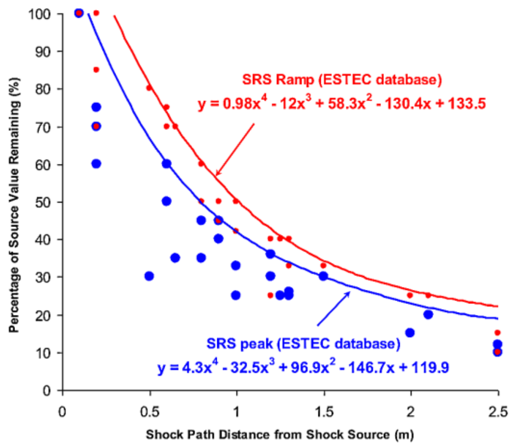


Figure 2-24: Comparison btw MIL-810-F (method 517) and ESTEC attenuation rules [13]

The pyroshock attenuation is a complex phenomenon, and for most cases the situations of shock propagation are different. The rules of thumb may not fit with every single pyroshock structure, but they can provide reliable guidelines. A suggestion of structural attenuation factors is given by Babuska et al. [25] The factors are shown in Table 2.3

Table 2.3: Typical structural attenuation factors [25]

Structural Feature	Attenuation (dB)
Rotational Joint	3
Distance	3
Right Angle	0
Bolted Joint	3
Bonded Joint	1
Flexure	0
Material	1

To get more convincible data, response accelerations at different positions on the shock table will be measured. The measurement will be done by changing the distance between measurement plate and excitation cylinder. The remaining percentage value of the SRS peak and the SRS ramp will be plotted, and the results will be compared with attenuation rules.

3. Results

In this chapter, the pyroshock response were calculated from different methods. The SRS curves of cantilever beam and fixed-pinned beams will be illustrated in section 3.1 to compare the FEA with numerical calculation using the beam equation. In section 3.2, the SRS of the shock table is displayed. The shock attenuation on the simplified shock table is presented in the form of remaining percentage of response level.

3.1. Comparison between FEA and numerical calculation applied to beams

The SRS curves of beams derived from FEA are compared with those from numerical calculation.

3.1.1. Materials

The material applied to both cantilever beam and fixed-pinned beam is Aluminum, and its important properties are $E = 71$ GPa, $\rho = 2770$ kg/m^3, $\nu = 0.33$.

3.1.2. Cantilever beam

In this section, three sets of geometry parameters will be applied to the cantilever beam. The length, width and the height of the beam are displayed in Table 3.1. The first natural frequency derived from beam equations are compared with the corresponding knee frequencies. The magnitudes of each set are also tabulated.

Table 3.1: Geometry parameters of cantilever beam

No.	Length (mm)	Width (mm)	Thickness (mm)	Frequencies (Hz)		Magnitude (G)
				Natural	Knee	
1	1 000	100	100	80.61	80	1 060
2	1 000	200	200	161.2	160	700
3	1 000	500	500	355.3	362	95
4	762	381	381	529.0	450	100
5	500	250	250	711.6	710	250
6	500	200	200	598.1	600	125
7	2 000	200	200	40.82	40	410

The Haversine force was added at the free end, and the response acceleration was measured at the same position. The SRS of the cantilever beams with different sizes are plotted in Figure 3.1 to 3.3. These figures show the comparison of the SRS from FE analysis and beam equation analysis. Figure 3.4 shows the comparison of results from FE analysis with different element types (solid elements and beam elements).

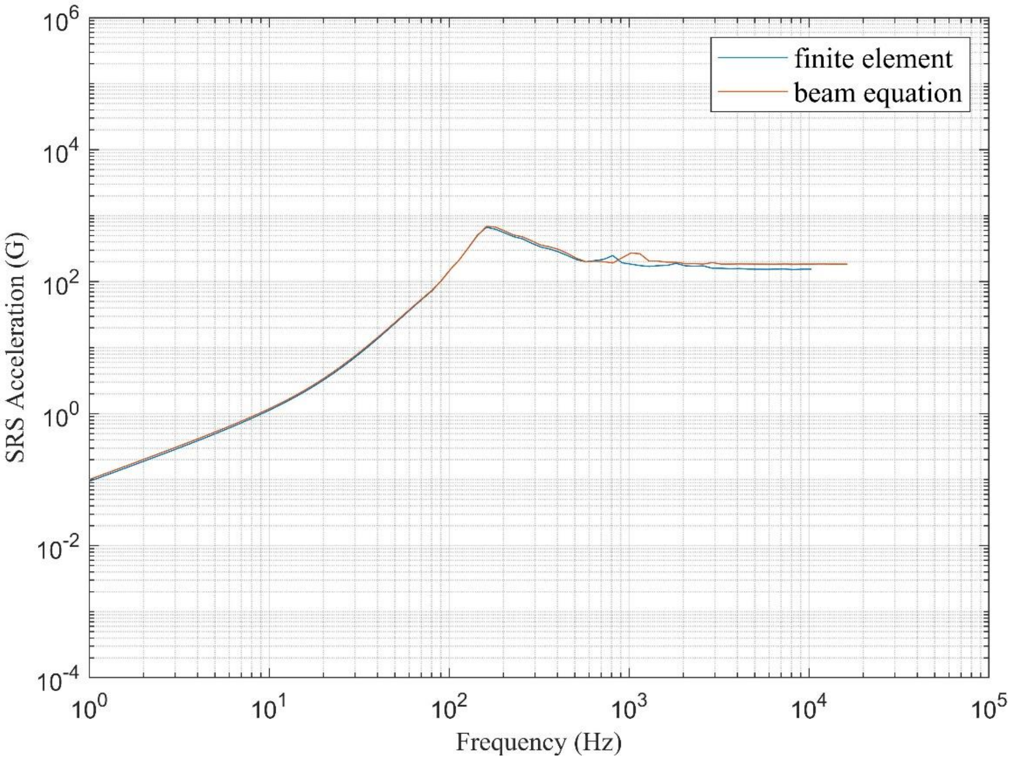


Figure 3.1: SRS of cantilever beam No.2 from FEA and beam equation

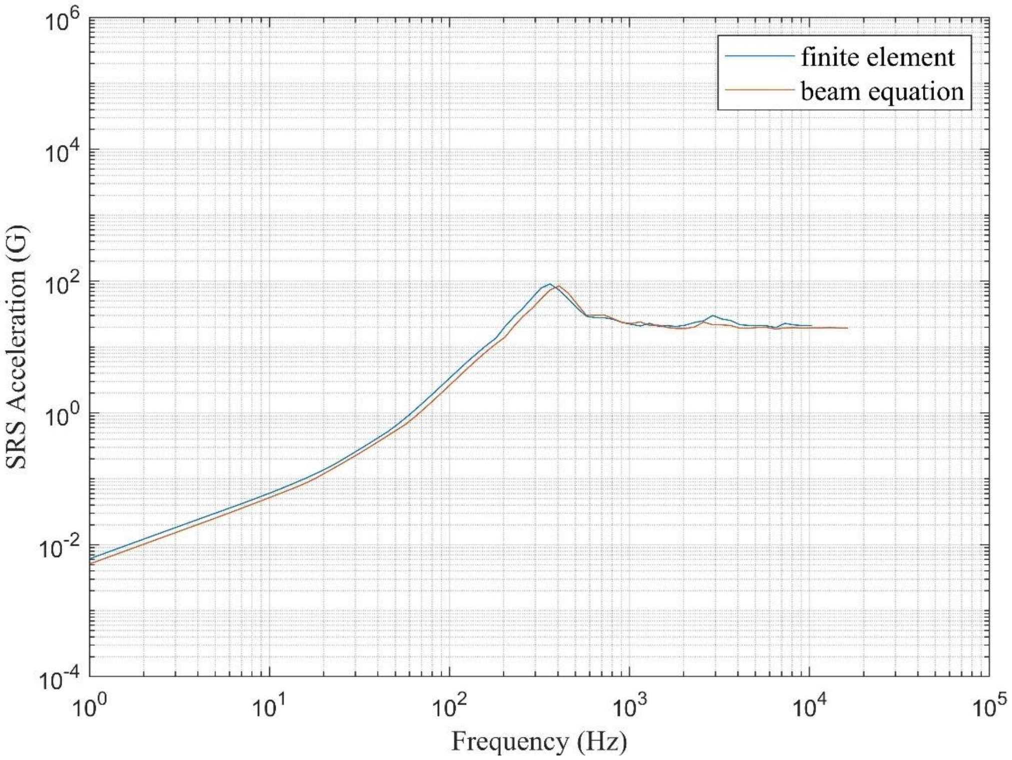


Figure 3.2: SRS of cantilever beam No.3 from FEA and beam equation

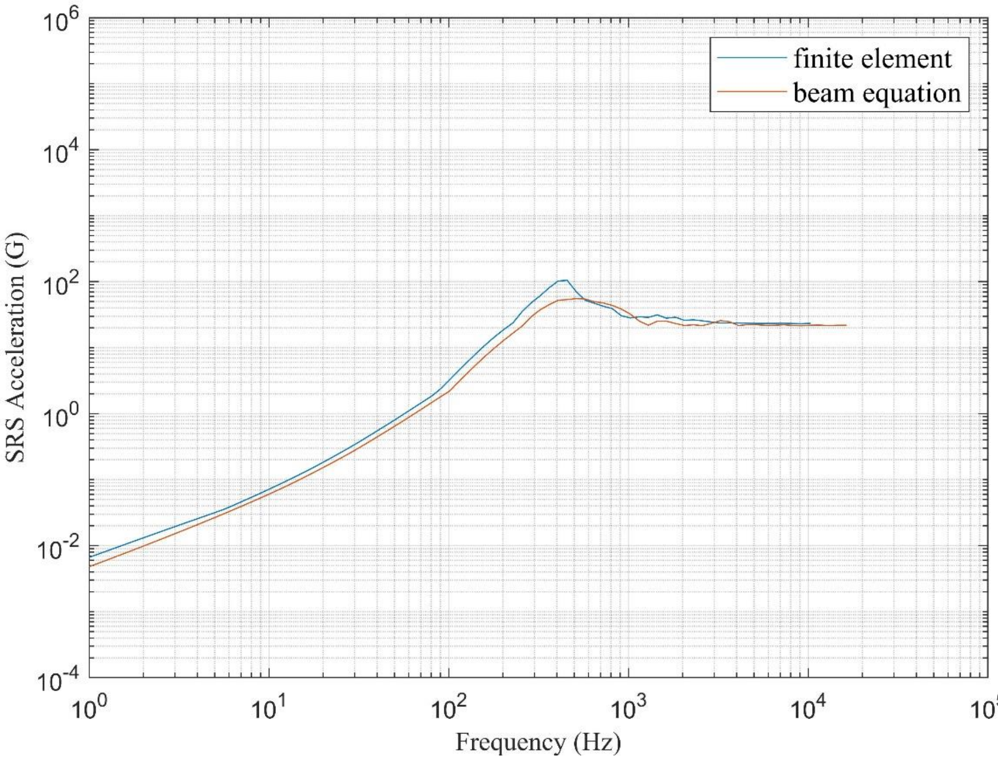


Figure 3.3: SRS of cantilever beam No.4 from FEA and beam equation

Figure 3.1 shows that for cantilever beam No.2, the SRS calculated from FEA is consistent with that from beam equation analysis in low frequency range. When the frequency is higher than 700 Hz, there is negligible difference between the two methods. Figure 3.2 shows that for beam No.3, the SRS curves from different analysis methods do not match equally: the response from FEA has a bit higher magnitude and a smaller knee frequency. Clear difference in both magnitudes and knee frequencies was observed for beam No.4 in Figure 3.3. For beams No.1 to 3, the magnitude decreased, while the knee frequency increased with increasing cross section areas. The SRS of beams No. 5, 6 and 7 are shown in Appendix 1 Figures 7.1 to 7.3.

From Table 3.1, it is clear that at fixed ratio of height/width to length, both the knee frequency and the magnitude decreased with the increase of table size (beams No. 3, 4 and 5). When both width and thickness were fixed, the knee frequency and decreased while the magnitude increased with increasing length (beams No. 2, 6 and 7).

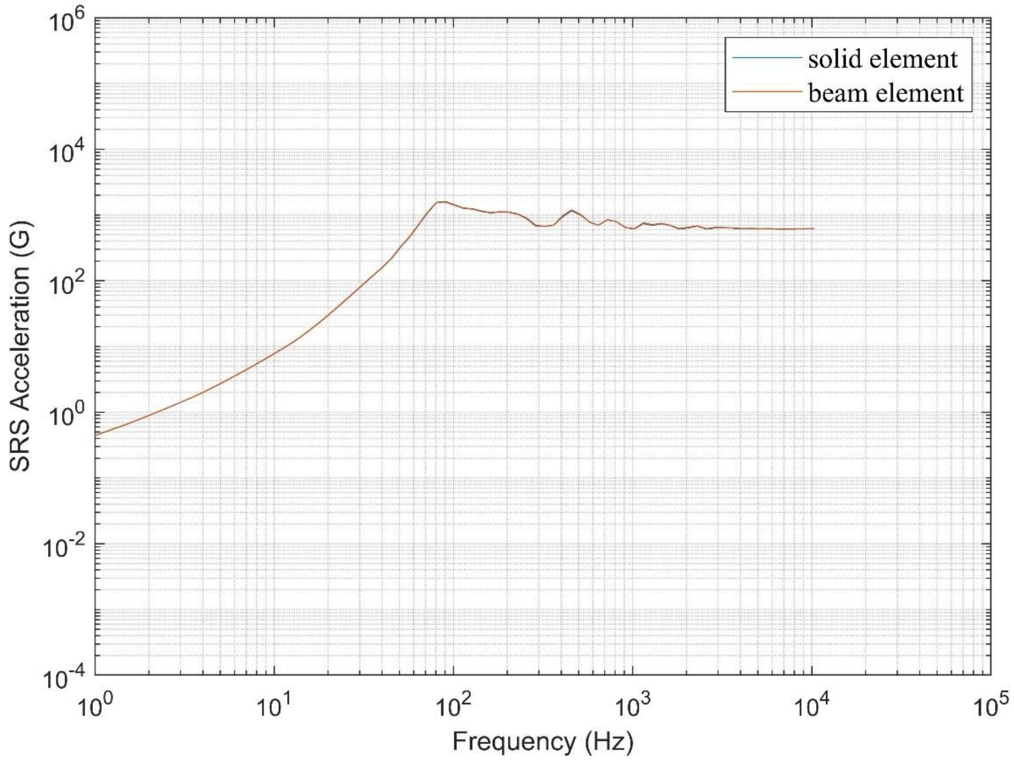


Figure 3.4: SRS curves of cantilever beam No.1 from solid elements and beam elements

Figure 3.4 displays the SRS of cantilever beam No.1 calculated from FEA with two different types of elements. The beam sets No.2 and No.3 were presented in Appendix 2 (Figures 7.16 and 7.17). According to these figures, it was known that the SRS calculated from FEA with different element types are almost the same.

3.1.3. Fixed-pinned beam

The SRS curves of fixed-pinned beams are shown in this section. The influence of width and thickness of the beams on the SRS is investigated. The beams had the same length of 1 000 mm, and they were divided into 3 groups: beams in group **1** had thicknesses varying from 10 mm to 450 mm, the widths were fixed; those in group **2** had fixed thicknesses, and their widths varied from 10 mm to 600 mm; the width and thickness of those in group **3** both varied. The parameters of the beams are displayed in Tables 3.2 to 3.4.

Table 3.2: Geometry parameters of fixed-pinned beam group **1**

No.	Length (mm)	Width (mm)	Thickness (mm)	Frequency (Hz)		Magnitude (G)
				Natural	Knee	
1	1 000	100	10	35.4	35	16 000
2	1 000	100	20	70.7	72	6 000
3	1 000	100	40	141.4	140	1 500
4	1 000	100	75	265.1	265	550
5	1 000	100	100	353.5	350	300
6	1 000	100	150	530.3	480	80
7	1 000	100	200	707.0	750	60
8	1 000	100	300	1 060.5	900	30
9	1 000	100	450	1 590.8	1200	13

The results of groups **1** and **2** were calculated from both FEA and beam equation, and the comparison between analysis methods were shown in Figures 3.5 to 3.9. Figure 3.10 shows the comparison between the SRS of group 3 from FEA simulation with different element types.

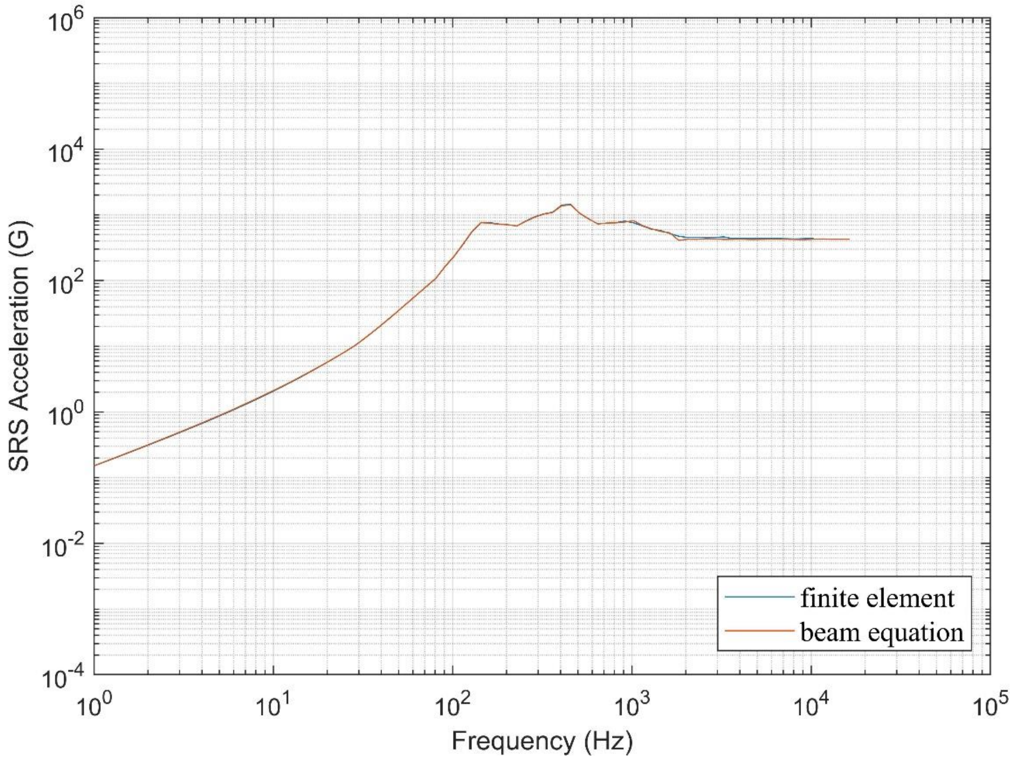


Figure 3.5: SRS of fixed-pinned beam No.3 from FEA and beam equation

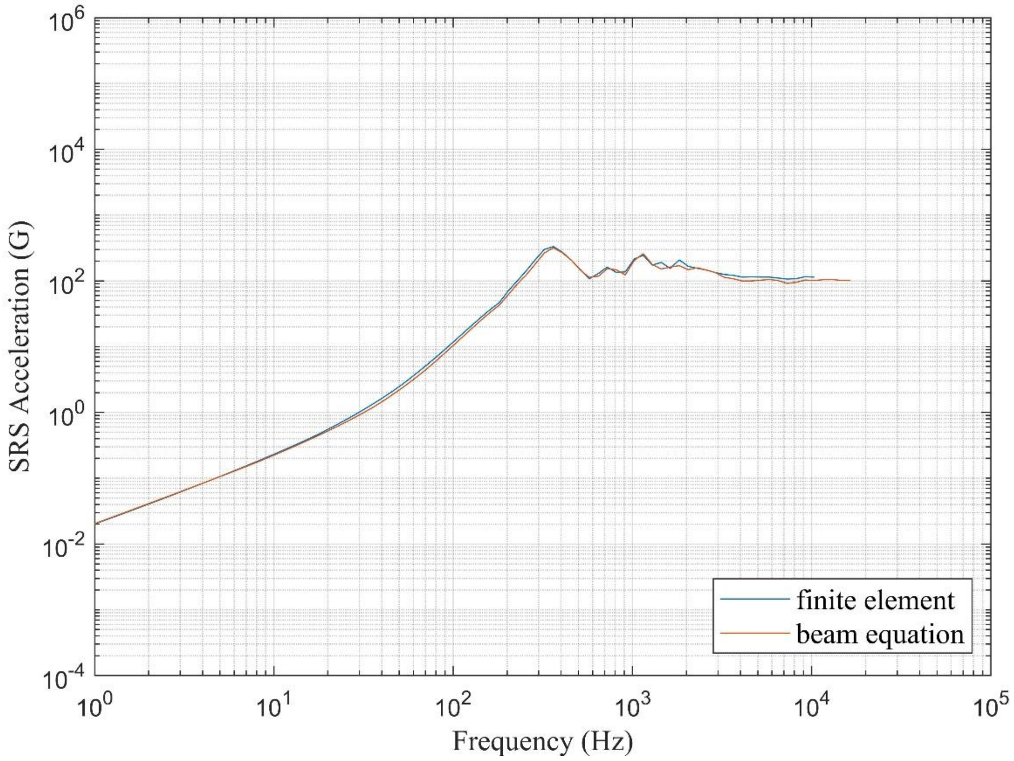


Figure 3.6: SRS of fixed-pinned beam No.5 from FEA and beam equation

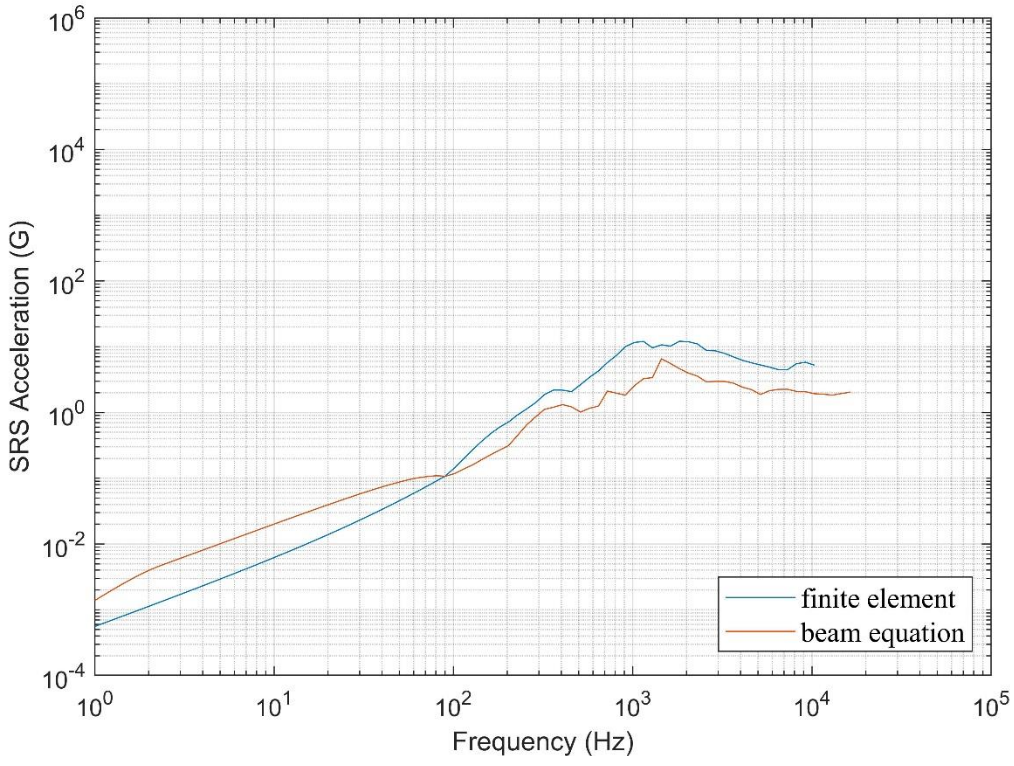


Figure 3.7: SRS of fixed-pinned beam No.9 from FEA and beam equation

Figures 3.5 demonstrates the comparison between the SRS of fixed-pinned beams No.3 calculated from two analysis methods. For beams No. 1-4, there is almost no difference between the SRS from two analysis methods (Figures 7.4 to 7.6). Then, with increasing thickness, the difference between the SRS from FEA and that from beam equation became more evident (Figure 3.6). When the ratio of thickness/length was larger than 0.1, the FEA results did not match well with beam equation results; further increasing this ratio, there occurred intersections between the SRS curves from two analysis methods (Figures 7.7 to 7.9). The knee and natural frequencies increased while magnitude decreased with increasing thickness (Table 3.3).

Table 3.3: Geometry parameters of fixed-pinned beam group **2**

No.	Length (mm)	Width (mm)	Height (mm)	Frequency (Hz)		Magnitude (G)
				Natural	Knee	
10	1 000	10	10	35.4	35	1.6e5
11	1 000	20	10	35.4	35	8e4
12	1 000	40	10	35.4	35	4e4
13	1 000	100	10	35.4	35	1.6e4
14	1 000	150	10	35.4	35	1e4
15	1 000	200	10	35.4	35	8 000
16	1 000	300	10	35.4	35	5 300
17	1 000	450	10	35.4	35	3 500
18	1 000	600	10	35.4	35	2 700

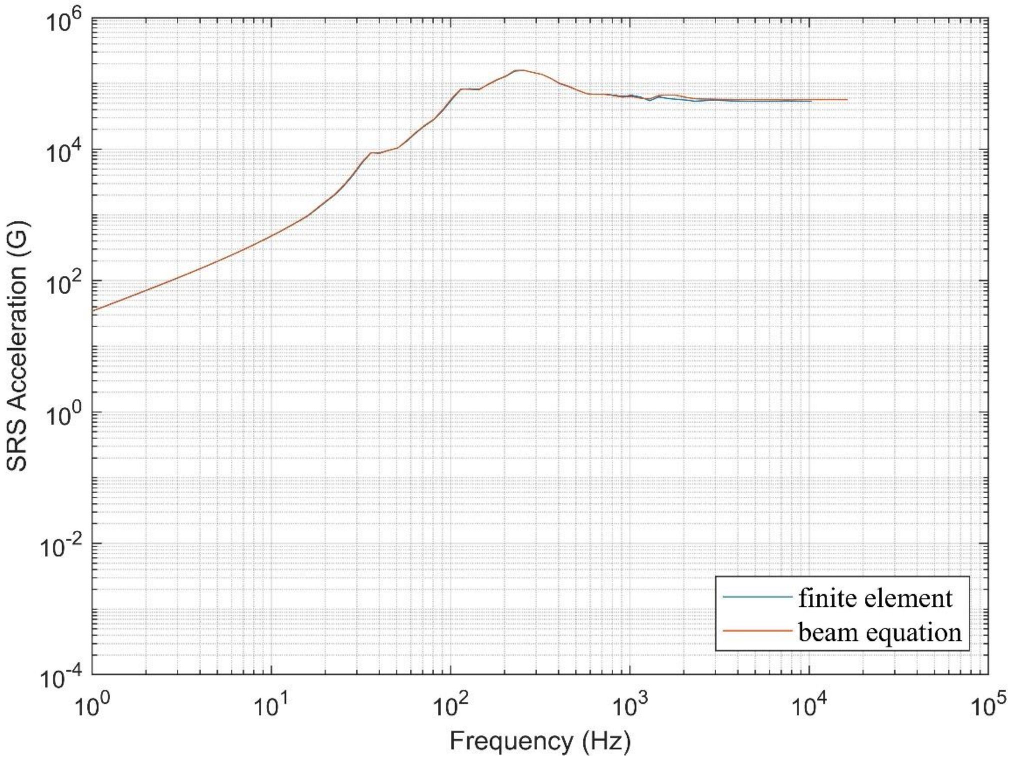


Figure: 3.8: SRS of fixed-pinned beam No.10 from FEA and beam equation

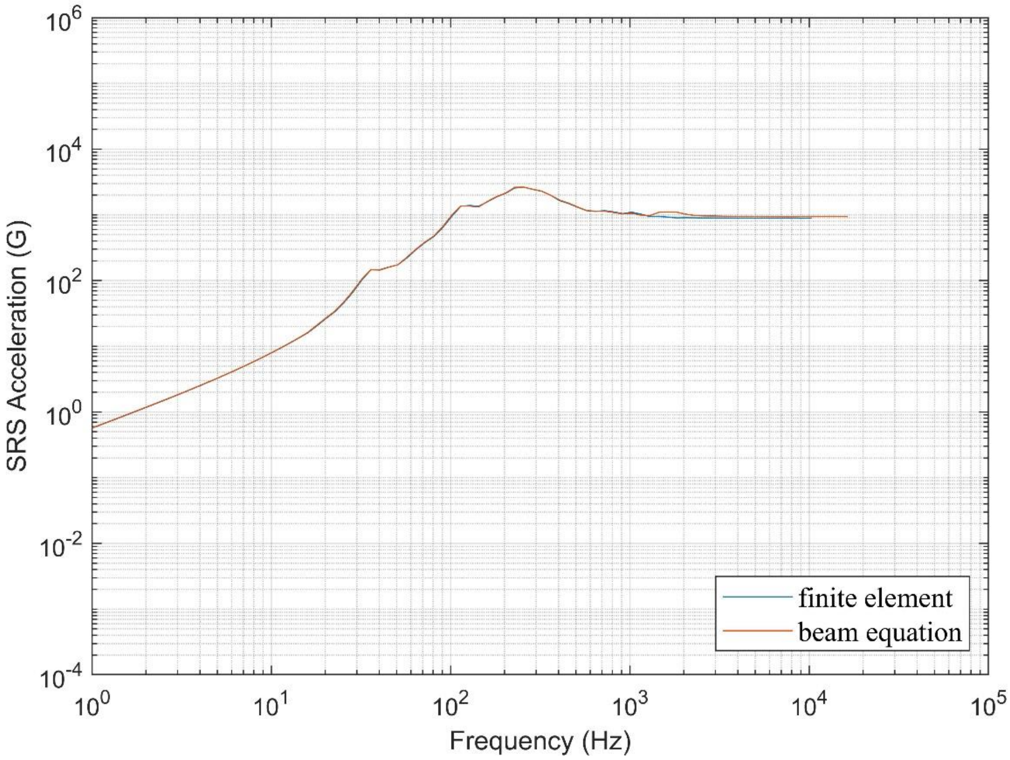


Figure 3.9: SRS of fixed-pinned beam No.18from FEA and beam equation

Figures 3.8 and 3.9 show the SRS of fixed-pinned beams No.10 and No.18 from two analysis methods, respectively. Those of beams No.11 to No 17 are shown in the Appendix 1 Figures 7.10 to 7.15. These figures reveal that there is almost no difference between two methods. Besides, the SRS curves of all beams in this group shares a same knee frequency, which reflects that all the beams in group **2** had the same first natural frequency. However, the magnitudes of these beams decrease with the increasing width.

Table 3.4: Geometry parameters of fixed-pinned beam group **3**

No.	Length (mm)	Width (mm)	Height (mm)	Frequency (Hz)	
				Natural	Knee
19	1 000	100	100	353.5	350
20	1 000	100	150	530.3	460
21	1 000	200	200	707.0	630
22	1 000	300	200	707.0	630
23	1 000	300	300	1 060.5	810

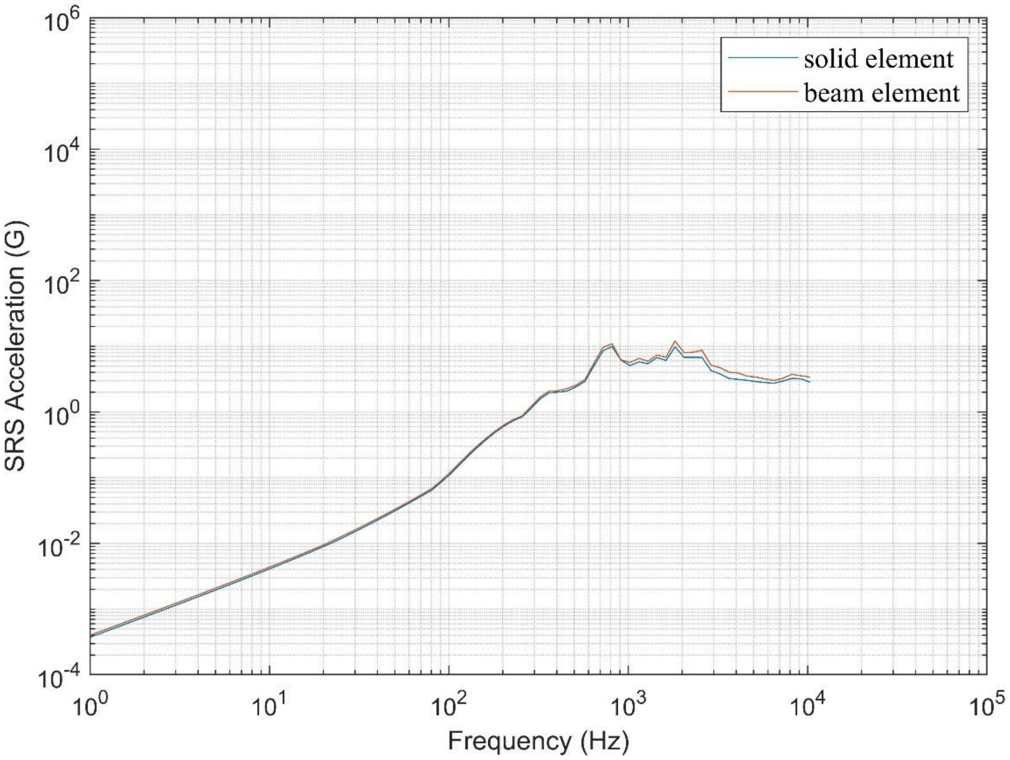


Figure 3.10: Comparison of SRS from solid elements and beam elements: SRS of fixed-pinned beam No.23

Figure 3.10 illustrates that the SRS curve from solid elements is a bit lower than that from beam elements in high frequency range, but the difference is neglectable. Similar results were found for other beams in this group (Figures 7.18 to 7.21 in Appendix 2).

3.2. Response of shock table and Attenuation

The investigation of the shock attenuation on the shock table is carried out with finite element analysis, as mentioned in section 2.5. In this section, the important parameters of the shock table and the material properties are tabulated. The response signal and the SRS of the shock table are plotted.

3.2.1. Geometry parameters

The geometry sizes of the shock table structure are presented here, including the size of the table, the height and the diameter of the supports, the thickness and the diameter of the damping materials and the plates. Two sets of table sizes were used, they shared the same geometry parameters except for length. The general view on the SRS, including the knee frequency and the SRS level are studied for both set **1** and **2**. The 2nd table is used to investigate the shock attenuation in structures.

The length, width and thickness of the table are shown in table 3.5.

Table 3.5: Size of the shock table

No.	Length (mm)	Width (mm)	Thickness (mm)
1	1 500	800	60
2	2 500	800	60

The sizes and positions of the supports are shown in Table 3.6.

Table 3.6: Size and position of the supports

No.	Diameter (mm)	Thickness (mm)	Distance to longer side (mm)	Distance to shorter side (mm)
1	50	30	150	250
2	50	30	150	250
3	50	30	400	150

The sizes and the positions of the excitation cylinder and the damping material **1** are shown in Table 3.7.

Table 3.7: Size and position of cylinder, plate and damping materials

	Diameter (mm)	Thickness (mm)	Distance to longer side (mm)	Distance to shorter side (mm)
Cylinder	33	12	400	300
Damping material 1	33	3	400	300

The sizes and the positions of the measurement plate and the damping material **2** are shown in Table 3.8.

Table 3.8: Size and position of cylinder, plate and damping materials

	Length × Width (mm)	Thickness (mm)	Distance to longer side (mm)	Distance to shorter side (mm)
Measurement plate	100 × 65	12	350	185
Damping material 2	100 × 33	3	350	200

3.2.2. Materials

The materials applied in the shock table structure are shown in table 3.9, and the properties of these materials are shown in table 3.10. A linear solver was employed, so the non-linear properties of the materials were excluded.

Table 3.9: Materials applied on different parts

Parts	Materials
Shock table	Aluminum Alloy
Supports	PolyRubberh
Measurement plate	Aluminum Alloy
Excitation cylinder	Aluminum Alloy
Damping materials	PolyRubberh

Table 3.10: Material properties

	Density (kg/mm^3)	Young's modulus (GPa)	Poison ratio
Aluminum Alloy	2770	71	0.33
PolyRubberh	1012	5	0.3

3.2.3. Position of the measurement points for attenuation study

The position of different measurement points on the shock table are displayed in Table 3.11. The 2nd set of shock table was used in this part. The measurement points are in an array in the midline along the longitudinal direction, so they are presented as the distance to the excitation point in *x*-direction.

Table 3.11: Distance from measurement points to the excitation point

No.	Distance in x-direction (mm)
1	0
2	150
3	300
4	450
5	600
6	750
7	900
8	1050
9	1200
10	1350
11	1500
12	1650
13	1800
14	1950
15	2100

3.2.4. Response and SRS of the shock table

The response acceleration and the SRS of the 1st shock table are shown in Figure 3.11 and 3.12.

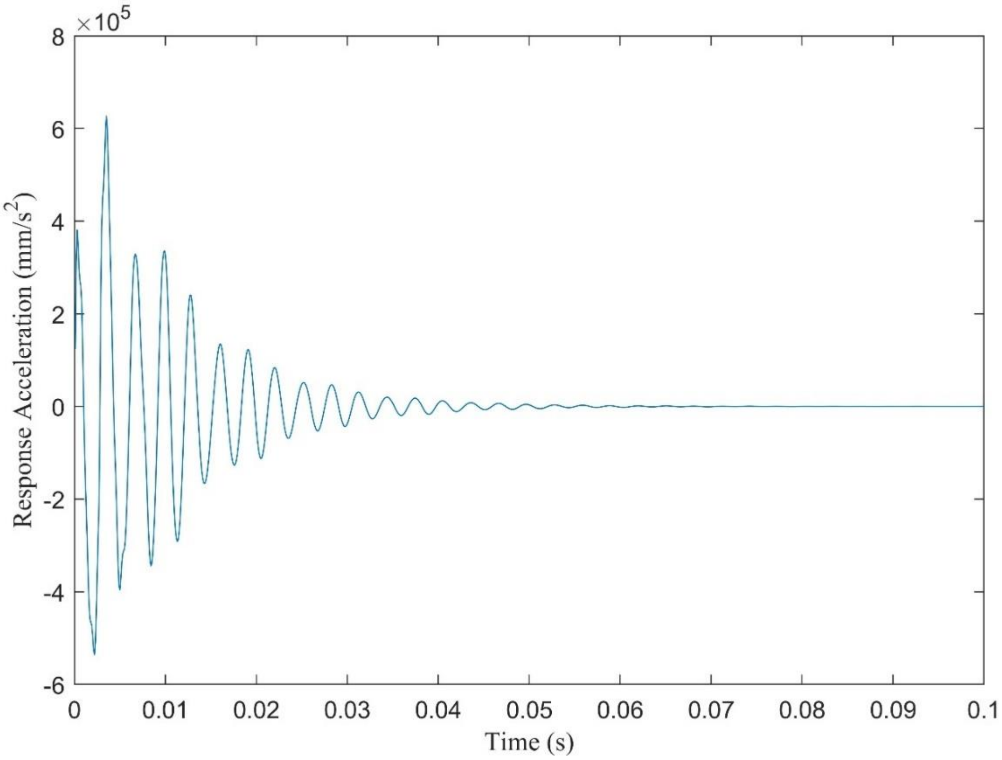


Figure 3.11: Response time-history of the 1st shock table

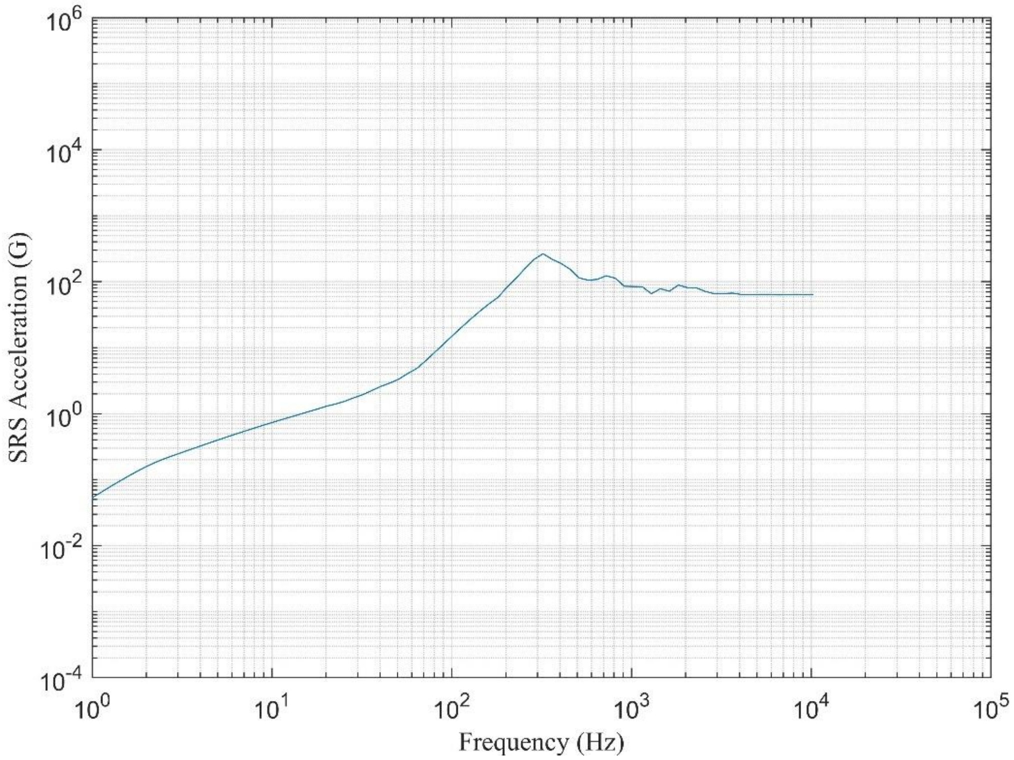


Figure 3.12: SRS of the 1st simplified shock table

Figure 3.11 shows that the response reaches a maximum value over 6×10^5 mm/s^2, and the duration is about 0.05 s. Figure 3.12 shows that the knee frequency of the SRS is about 320 Hz, and the peak value reaches 260G. The SRS plateaus at about 60G.

The response acceleration and the SRS of the 2nd shock table are shown in Figure 3.13 and 3.14.

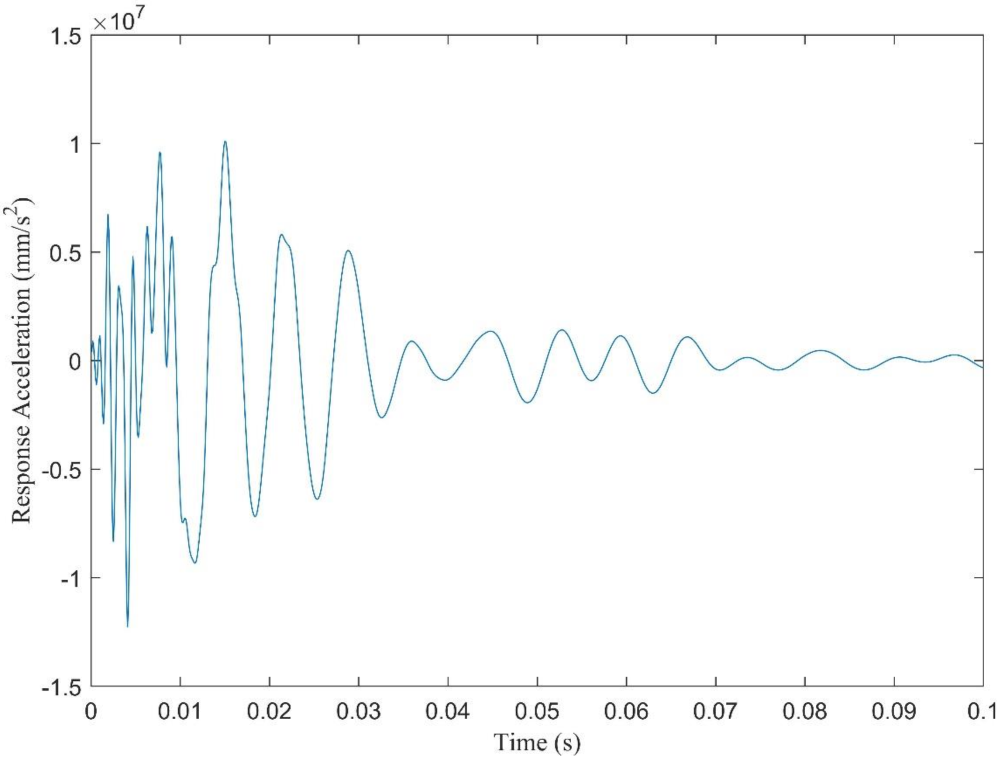


Figure 3.13: Response time-history of the 2nd shock table

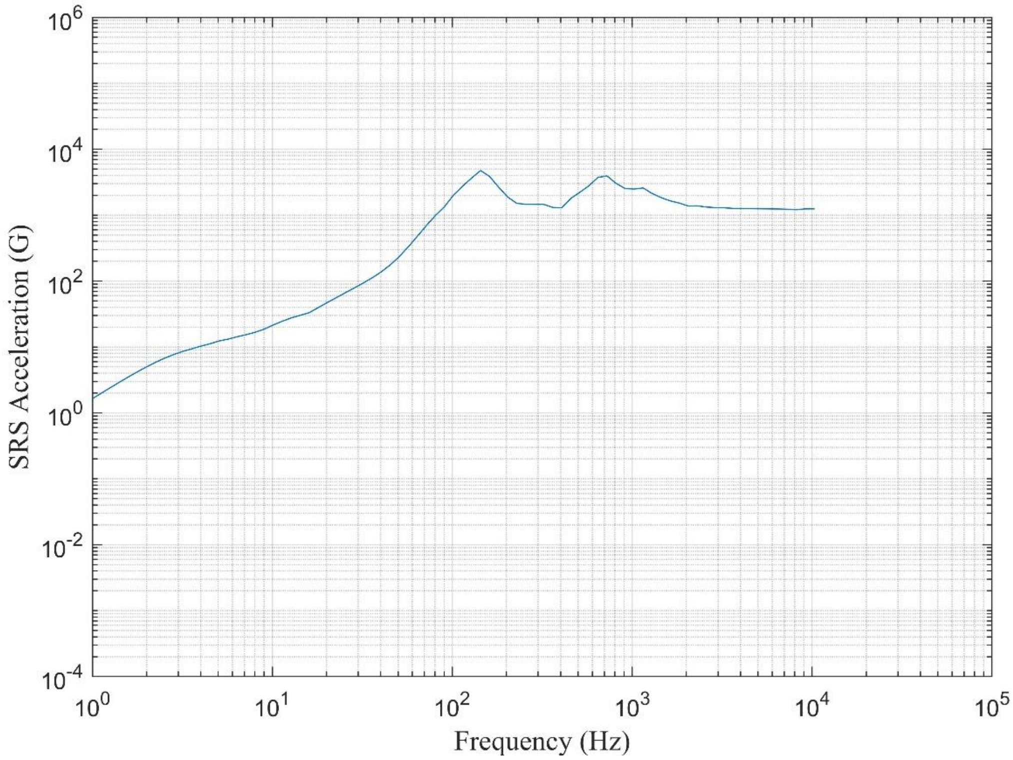


Figure 3.14: SRS of the 2nd simplified shock table

Figure 3.13 shows that the maximum absolute value of time-history response signal is over $1 \times 10^7\ \mathrm{mm/s^2}$, the acceleration did not fully decay after 0.1 second. Figure 3.14 shows that the knee frequency of the 2nd shock table is about 150 Hz. The maximum value is around 4900G, and the plateau is about 1200G in frequency range from 2 kHz to 10 kHz. Similar to beams, the knee frequency decreased while the magnitude increased with increasing length of stock table.

3.2.5. Attenuation

In this section the shock responses at different positions on the shock table are measured. The attenuation is described by the remaining SRS value compared to the value at excitation point (pyroshock source). Two frequencies in the SRS are chosen as the reference of attenuation, one is on the ramp of the curve, and the other is the knee frequency. Their attenuation regulars are called ramp attenuation and peak attenuation respectively. Figure 3.13 shows the comparison between simulation results and ESTEC rules.

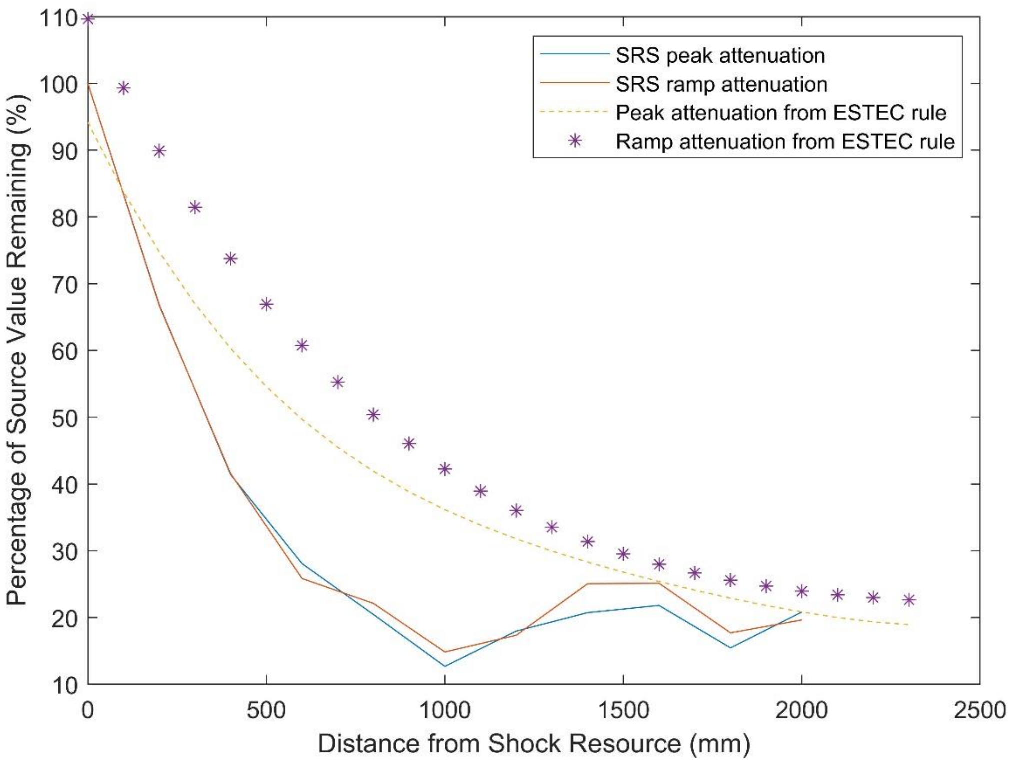


Figure 3.15: the attenuation in the simplified shock table due to distance

Figure 3.15 shows the variation of the remaining values of SRS depends on distance. The remaining value drops with increasing distance. However, fluctuations with values of about 10% to 30% appear in the distance range from 1000 mm to 2150 mm.

The peak attenuation matches well with the ramp attenuation when the distance is less than 400 mm. From 400 mm to 1000 mm, the simulation results drop much more rapidly than ESTEC rules. When the distance is above 1300 mm, the remaining values of simulation results and the rules gradually tend to be consistent with each other and they both get towards 20%.

4. Discussion

4.1. Comparison between beam equation analysis and finite element analysis

In FEA simulation, more factors including motions and bending are involved. However, only bending is considered in the beam equation. Therefore, there could be difference between the results from the two methods in some cases.

4.1.1. Cantilever beams

In the study on cantilever beams, the ratio of thickness/width was kept the same, so the discussion is based on the relationship between cross section and the length. When the ratio of 'cross section/length' of the beams were relatively small, the SRS from FEA simulation matched well with those from beam equation. But the results from different analysis methods did not agree with each other at relatively large cross section (the ratio of width/length equal to 1/2 in this thesis).

A possible reason for this phenomenon is that the beam equation is based on Bernoulli's beam for bending, waves caused by other motions were thus ignored in the calculation. Therefore, it can be assumed that if the cross sections of the beams are sufficiently small, the beam equation analysis could give accurate results.

4.1.2. Fixed-pinned beams

For fixed-pinned beams, the influence of width and thickness was investigated. For group 1, the length and width were fixed. Difference between two analysis methods occurred when the ratio of thickness/length is larger than 1/10. The differences in both knee frequency and magnitude enlarged with increasing thickness. However, for group 2, the length and thickness were fixed, and the thickness was set relatively small (thickness/length=1/100), and there shows no difference between analysis methods was observed.

Therefore, the beam equation analysis seems accurate only when beams are thin, and the width does not influence the accuracy of the analysis. A possible explanation is that, when a beam is thick, vibration caused by motions other than bending, such as torsion and reflection of waves, will take more portion in shock response. These motions are considered in the FEA, but not in Bernoulli beam equation, which caused the occurrence of difference.

4.2. Comparison between solid elements and beam elements

For both cantilever beams and fixed-pinned beams, the results from solid elements and beam elements are basically consistent with each other. The sizes of beams and the ratio between dimensions do not matter very much. Therefore, the solid elements are proved to be applicable in FEA on shock response simulation.

4.3. SRS features

The influence of beam geometry on the SRS was investigated. According to the simulation results, for most cases the knee frequencies of the SRS appear near the first natural frequency of the beams. Based on that, it makes sense to believe that the knee frequency of the SRS is equal to the first natural frequency of a structure.

Under fixed thickness and length, the knee frequency and the shape of the SRS does not change with varying width; when length and width are kept constant, the knee frequency increases with rising thickness; whereas it decreases with increasing length when width and thickness are unchanged. Therefore, it can be concluded that the first natural frequency of a beam is dependent on the thickness and the length, but generally independent on the width. However, the width exerts influence on the SRS magnitude: increasing width leads to decreasing response levels. The results can provide some ideas for controlling the shock response level and keeping the knee frequency and curve shape constant by modifying the geometry dimensions of complicated structures.

4.3. Response of shock table and attenuation

The simulation on both 1st and 2nd shock tables provided a general view on shock response behaviors of space structures. The first natural frequency of this structure can be estimated from the SRS. According to the studies on beam response, changing the width of the table can vary the SRS level without moving knee frequency; altering the length or the thickness can adjust both knee frequency and level of the SRS.

The influence of the distance on shock level was studied on the 2nd shock table. The shock attenuation was presented as remaining percentage of the SRS from shock source. The simulation result had a similar trend with ESTEC attenuation rules although there was difference between them. The difference might be caused by the following reasons:

1. The simplified model for simulation has removed some details of the geometry including joints. The joints in structures also play important roles in exhausting vibration energy, so the lack of some details may bring errors.

2. The modal transient solver neglected non-linear features, so the simulation was not exactly the same as the real situation.

3. In real shock environment, localized reflection and nonideal contact between surfaces will bring in uncertainty, which affects the experimental results on the other hand.

4. The attenuation rules were derived from empirical data, they would not fit with every specific case.

In conclusion, from the analysis results in this study, the penetration of pyroshock in an aerospace structure can be determined by geometry features of the structure and the distance from shock source. According to attenuation rule of thumb, the joints in the structure also has influence on the pyroshock penetration. To avoid damage on vibration-sensitive device, following suggestions are given based on this study:

1. The knee frequency of the SRS is related to the natural frequency of a structure.

2. Changing surface area can adjust the SRS level of a structure.

3. The device should be placed as far from the shock source as possible.

4. The influence of joints and material properties were not investigated in this thesis. However, they could be considered as important factors for decreasing the SRS levels according to different criteria and references.

5. Conclusion

The shock response spectra were calculated by a recursive digital filter method. Theoretical calculation of shock response on beams was executed based on a combination of Bernoulli beam vibration equation in bending and principle of virtual work. The shock response vibration data on beams obtained from FEA models was compared with that from theoretical calculations. The results demonstrated that FEA models are valid for pyroshock response simulations of beams. No difference was found between the SRS simulated by FEA with solid elements and with beam elements.

1. For cantilever beams, the difference between the SRS from FEA and from beam equation increase with increasing ratio of cross area/length. Under fixed cross section area, the knee frequency decreased, while the magnitude increased with increasing length; when the ratio of height/width/length kept constant, both knee frequency and the magnitude decreased with table size; at fixed length and ratio of height/width, the knee frequency increased, whereas the magnitude decreased with increasing cross area.

2. In the case of fixed-pinned beams, at fixed length and width, the magnitude decreased, while the knee frequency increased and difference between FEA and beam equation become larger with increasing thickness. When the length and thickness were kept unchanged, the magnitude decreased, whereas the knee frequency kept the same with increasing width. In addition, no difference between FEA and beam equation results was observed in the latter case.

3. The natural frequency and the shock response of the shock table is estimated with an FE model. At fixed width and thickness, the knee frequency decreased, while the magnitude increased with increasing length. The curve acquired from simulation roughly agreed with that from ESTEC attenuation rules. However, clear difference between simulation and ESTEC rule was found at distance between 400-1300 mm. When distance is larger than 1300 mm, they tend to be consistent with each other and get close to 20%.

6. References

[1] M. N. Ichchou and O. Bareille (2012). *A discussion on high-frequency wave energy behavior in viscoelastic media*, Journal of Engineering Mathematics, 75: 127–138.

[2] ECSS-E-HB-32-26A (2013), *Space Engineering, Spacecraft Mechanical Loads Analysis Handbook*, February 19, 2013.

[3] A. Brandt and R. Singhal (2016), *Shock & Vibration, Aircraft/Aerospace, Energy Harvesting, Acoustics & Optics, Volume 9*, Proceedings of the 34th IMAC, A Conference and Exposition on Structural Dynamics: 29–52.

[4] S. Quan, T. Juan and M. Qiang (2009), *Damage effect analysis and experiment for electronic equipment in impact vibration environment*, Dept. of Management Engineering, Shijiazhuang Mechanical Engineering Coll., Shijiazhuang 050003, P. R. China Journal of Systems Engineering and Electronics 20(6): Pages 1384–1388.

[5] O. G. Demenko and A. S. Birukov (2018), *The Effect on Small Spacecraft of the Shock Loading Level by Actuation of the Separation System*, Solar System Research, 52(7): 655–661.

[6] A. G. Pérez and F. S. Palmer, G. Alonso and A. Ravanbakhsh (2018), *Overview and application of FEA methods for shock analysis in space instruments*, Aerospace Science and Technology 80: 572–586.

[7] O. M. F. Morais and C. M. A. Vasques (2017), *Shock environment design for space equipment testing*, Proc IMechE Part G: Journal of Aerospace Engineering 2017, 231(6): 1154–1167.

[8] B. I. Kunicka (2017), *Spacecraft dynamic analysis and correlation with test results*, Degree project, Department of Computer Science, Electrical and Space Engineering, Luleå University of Technology.

[9] F. Ding, W. Wan, T. Mao, M. Wang, B. Ning, B. Zha and B. Xiong (2014), *Ionospheric response to the shock and acoustic waves excited by the launch of the Shenzhou 10 spacecraft*, Geophysical Research Letters, 41: 3351–3358, doi:10.1002/2014GL060107.

[10] M. S. Ahmed, M. Ghommem and A. Abdelkefi (2018), *Shock response of electrostatically coupled microbeams under the squeeze-film damping effect*, Acta Mechaica, 229: 5051–5065.

[11] H. Niwa, M. Warashina, U. Watanabe and T. Tanoue (2012), *Vibration isolation system to reduce high frequency vibrations due to airplane crash*, ASME 2012 Power Conference. July 30 - August 3, 2012, Anaheim, California, USA.

[12] A. George, A. Sunny and J. Cyriac (2017), *Shock Response Spectrum of Launch Vehicle Using LabVIEW*. 2017 International Conference on Innovations in information Embedded and Communication Systems.

[13] ECSS-E-HB-32-25A (2015), *Space Engineering, Mechanical Shock Design and Verification Handbook*. July 14, 2015.

[14] NASA-STD-7003 (1999), *Pyroshock Test Criteria*, May 18, 1999.

[15] MIL-STD-810F (2000), *Environmental Engineering considerations and laboratory tests*, MIL Technical Standard, Jan. 2000.

[16] D.L. Van Ert, *Survey of pyroshock prediction methodology*.

[17] S. Grahn (2019), *CubeSat Proposal, MIST- Miniature Student Satellite*, KTH Space Center. Stockholm: 19–26.

[18] NASA-HDBK-7005 (2001), *Dynamic Environment Criteria*, March 13, 2001.

[19] J. Wijker (2004), *Mechanical Vibrations in Spacecraft Design*, Springer-VerlagBerlin Heidelberg, New York. 2004: 19–42, 173–200.

[20] J. Tuma and P. Koci (2011), *Calculation of a shock response spectrum*, May 2011, Acta Montanistica Slovaca 16(1): 404–409.

[21] K. Ahlin, M. Magnevall and A. Josefsson (2006), *Simulation of forced response in linear and nonlinear mechanical systems using digital filters*,

http://www.bth.se/tek/amt/nonlinear.nsf/(WebFiles)/F75E7709B9C6F47DC12571D50033078C/$FILE/ISMA_2006_444.pdf

[22] T. Irvine (2012), *A Digital Recursive Filtering Method for Calculating the Base Input for a Measured Response Acceleration*, March 28, 2012.

[23] D. Thorby (2008), *Structural Dynamics and Vibration in Practice-An Engineering Handbook*, Chapter 8 Vibration of structures: 181–186

[24] T. Irvine (2013). *The Response of a Cantilever Beam Subjected to an Applied Concentrated Half-sine Force Pulse*, Revision B. January 21, 2013.

http://www.vibrationdata.com/beams.htm

[25] V. Babuška[1], S. Gomez[2], S. Smith[3], C. Hammetter[4] and D. Murphy[5] (2016), *Spacecraft Pyroshock Attenuation in Three Parts*, American Institute of Aeronautics and Astronautics. SAND2016-12415C.

Appendix 1: Comparison between beam equation and finite element analysis

Cantilever beams

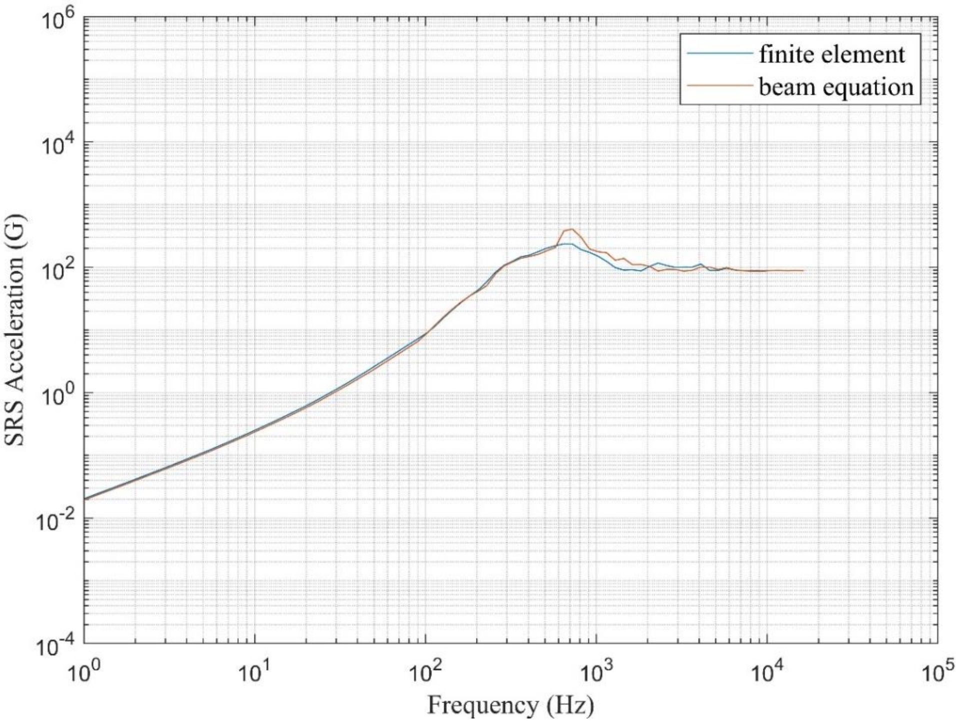


Figure 7.1: Comparison of SRS of cantilever beam No.5 from FEA and beam equation

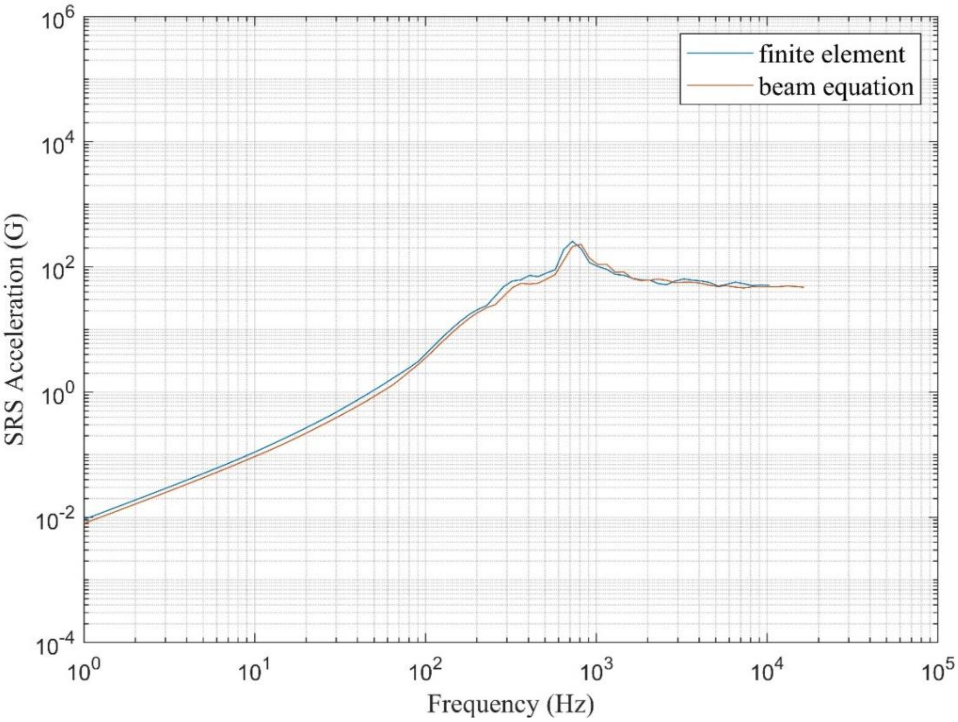


Figure 7.2: Comparison of SRS of cantilever beam No.6 from FEA and beam equation

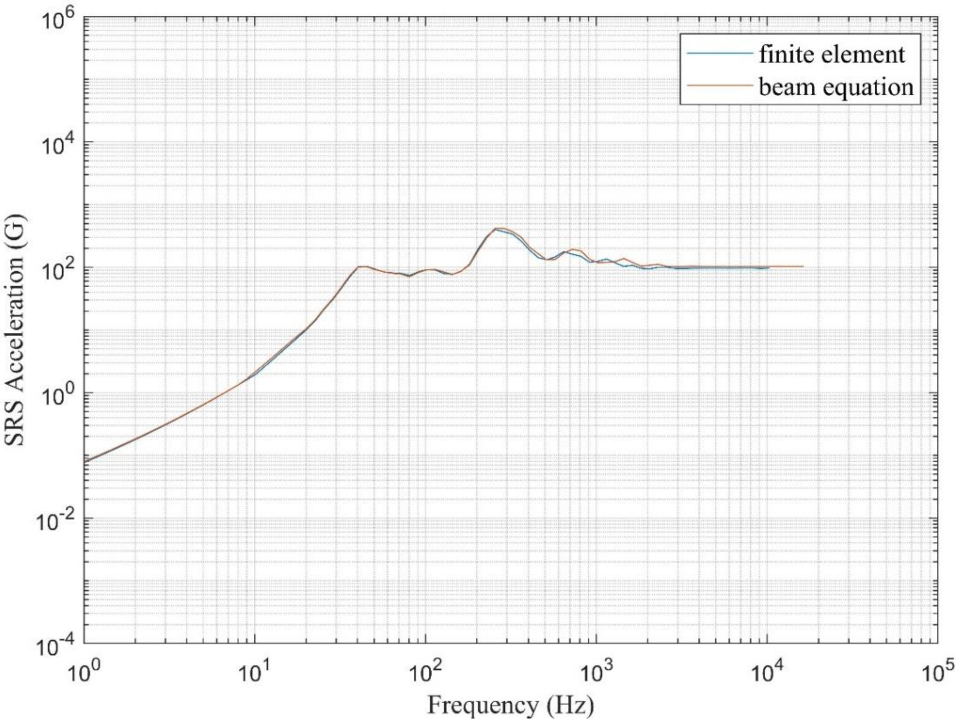


Figure 7.3: Comparison of SRS of cantilever beam No.7 from FEA and beam equation

Fixed-pinned beams group 1

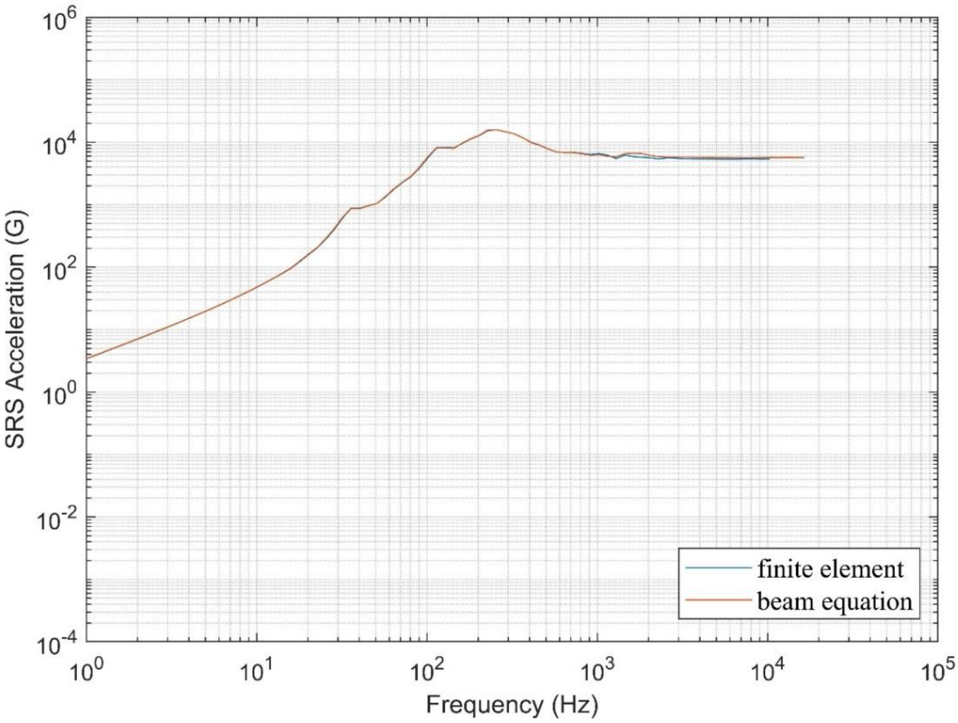


Figure 7.4: Comparison of SRS of fixed-pinned beam No.1 from FEA and beam equation

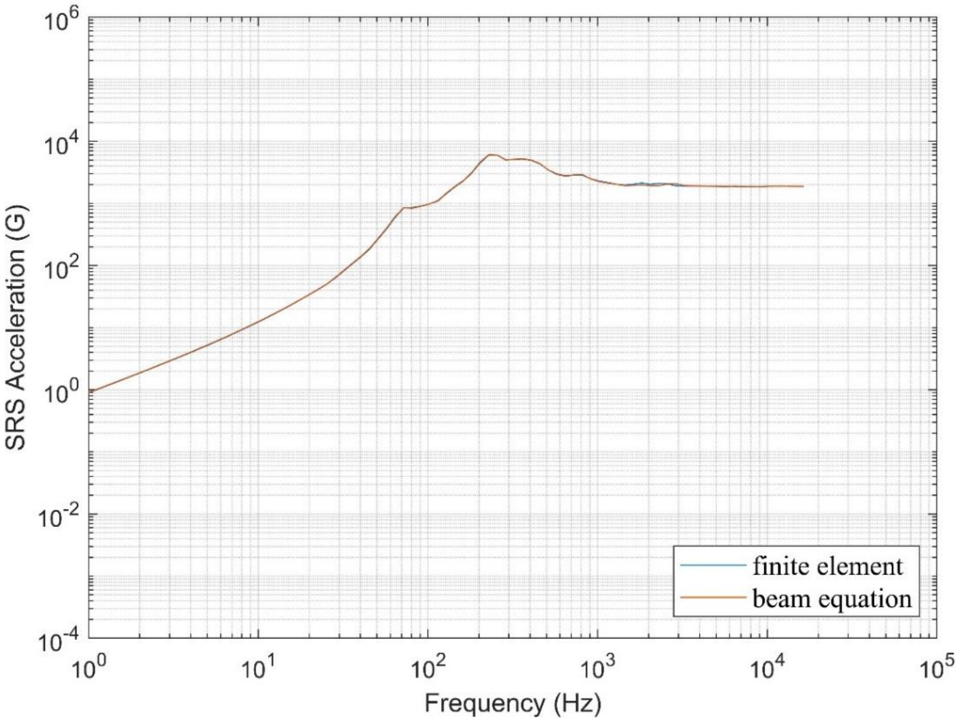


Figure 7.5: Comparison of SRS of fixed-pinned beam No.2 from FEA and beam equation

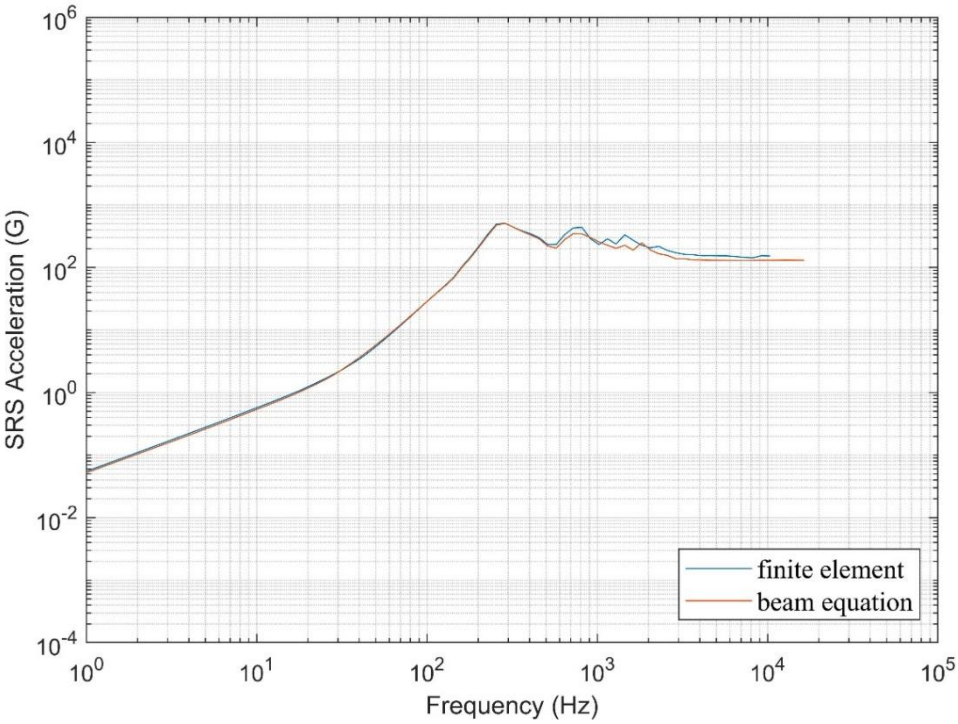


Figure 7.6: Comparison of SRS of fixed-pinned beam No.4 from FEA and beam equation

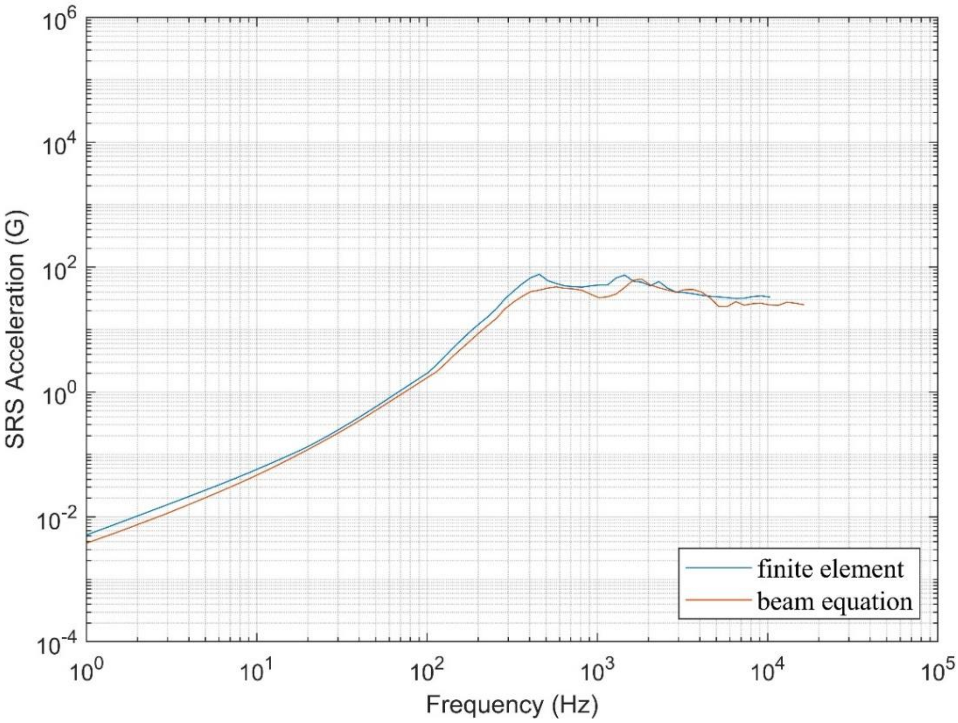


Figure 7.7: Comparison of SRS of fixed-pinned beam No.6 from FEA and beam equation

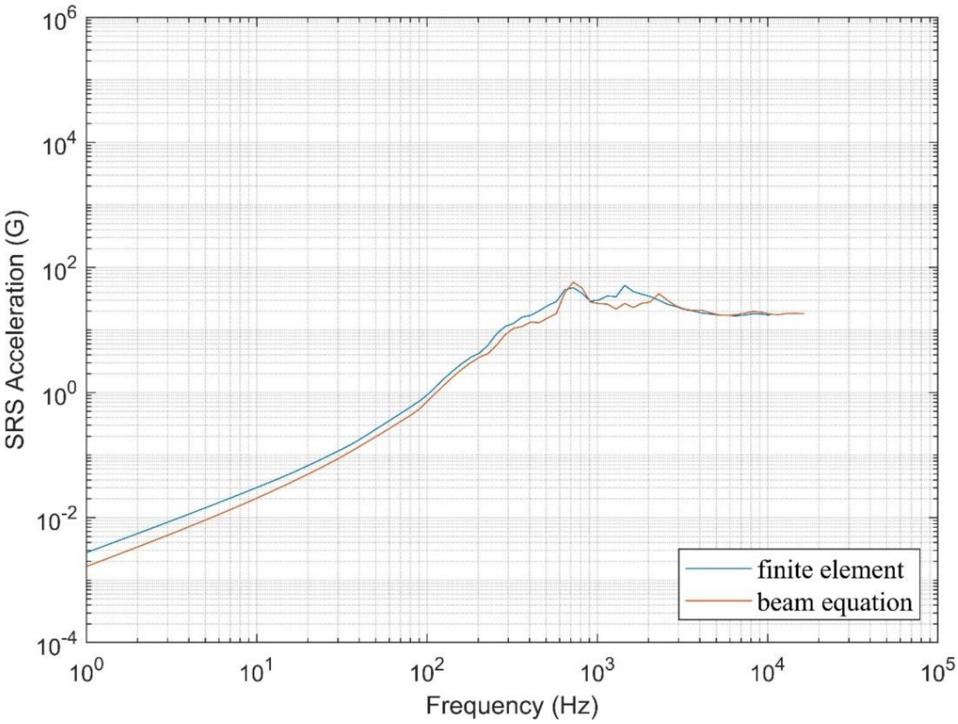


Figure 7.8: Comparison of SRS of fixed-pinned beam No.7 from FEA and beam equation

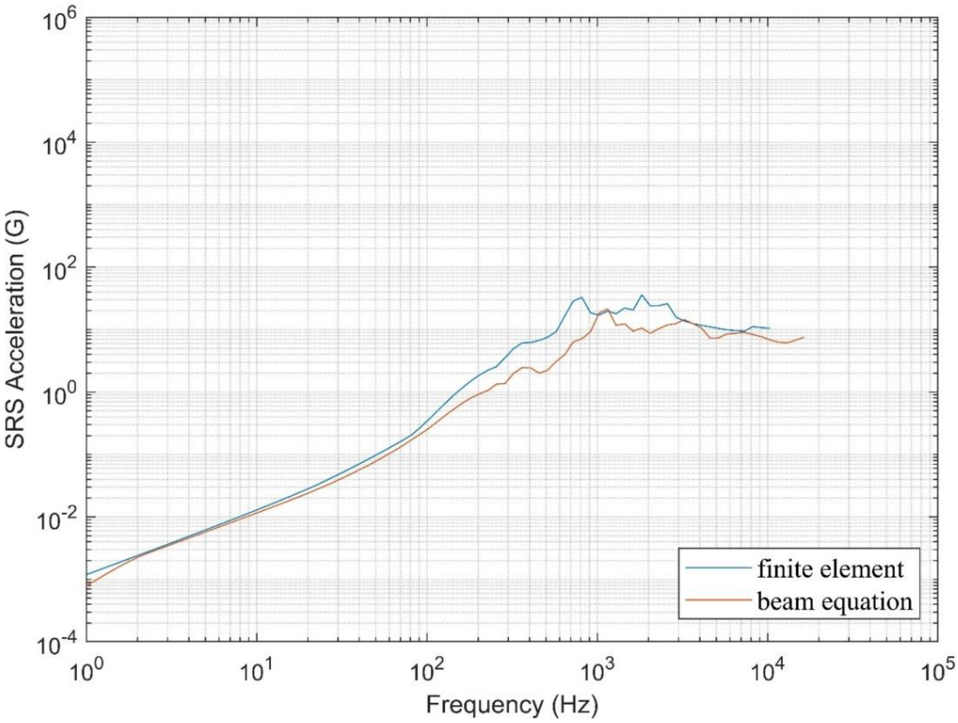


Figure 7.9: Comparison of SRS of fixed-pinned beam No.8 from FEA and beam equation

Fixed-pinned beams group 2

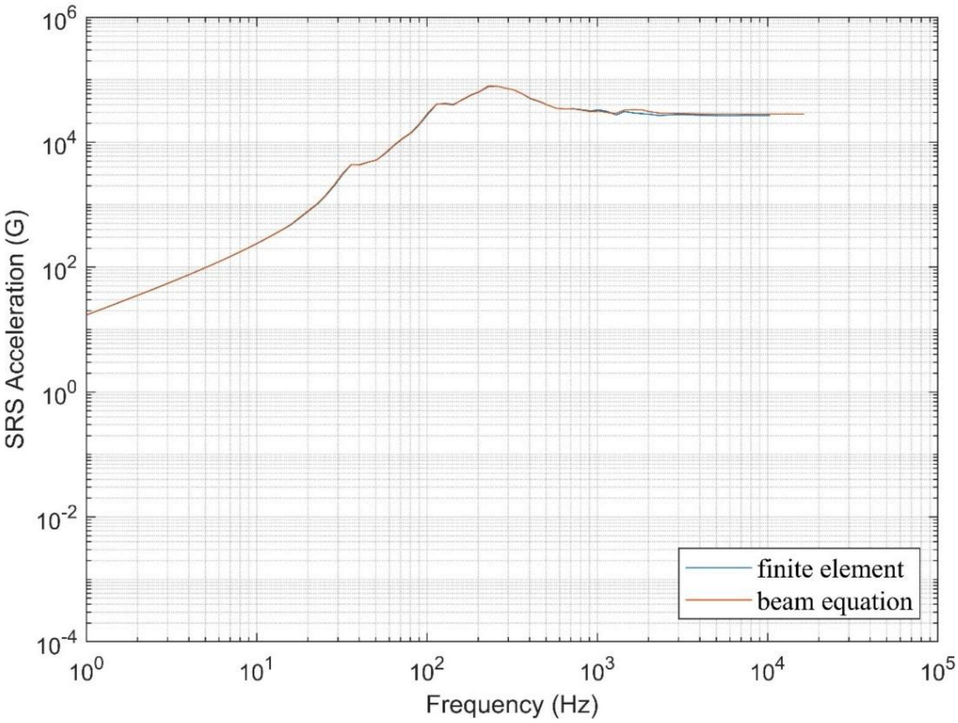


Figure 7.10: Comparison of SRS of fixed-pinned beam No.11 from FEA and beam equation

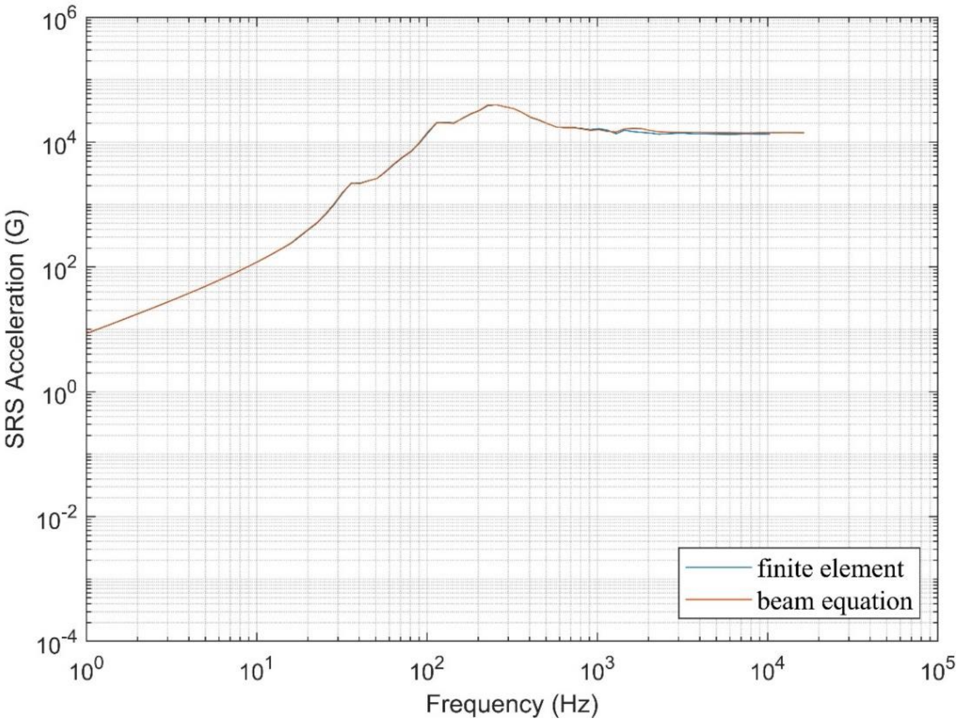


Figure 7.11: Comparison of SRS of fixed-pinned beam No.12 from FEA and beam equation

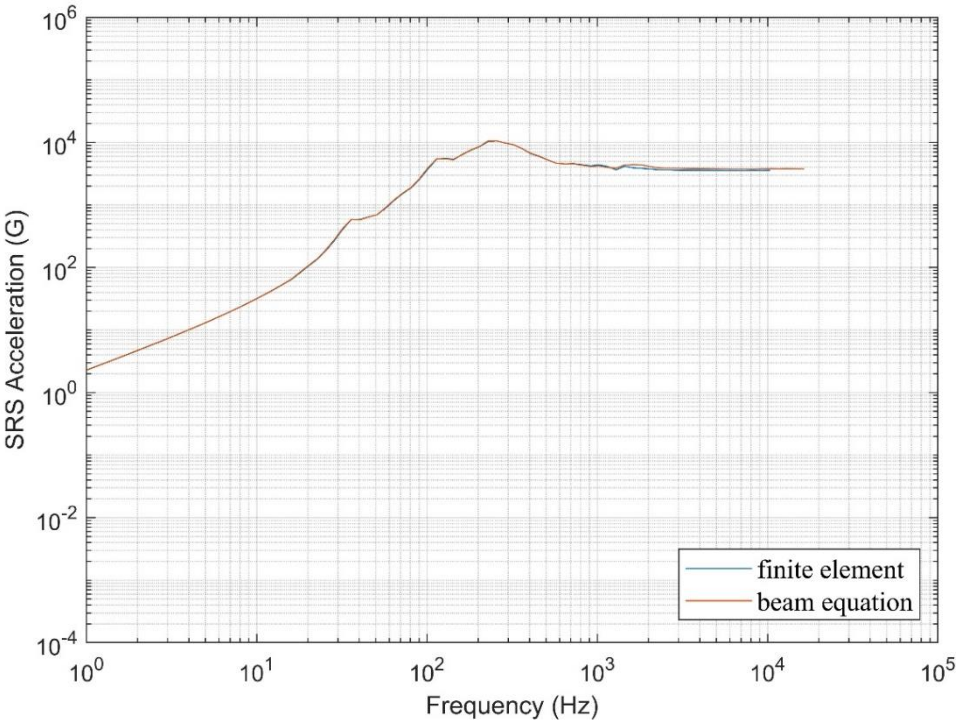


Figure 7.12: Comparison of SRS of fixed-pinned beam No.14 from FEA and beam equation

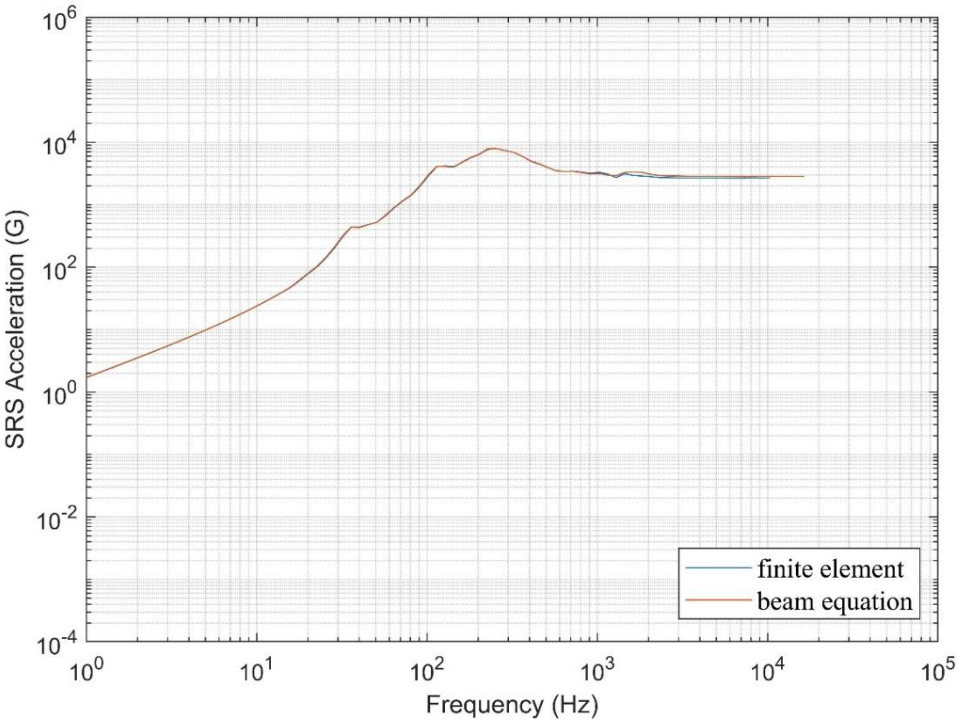


Figure 7.13: Comparison of SRS of fixed-pinned beam No.15 from FEA and beam equation

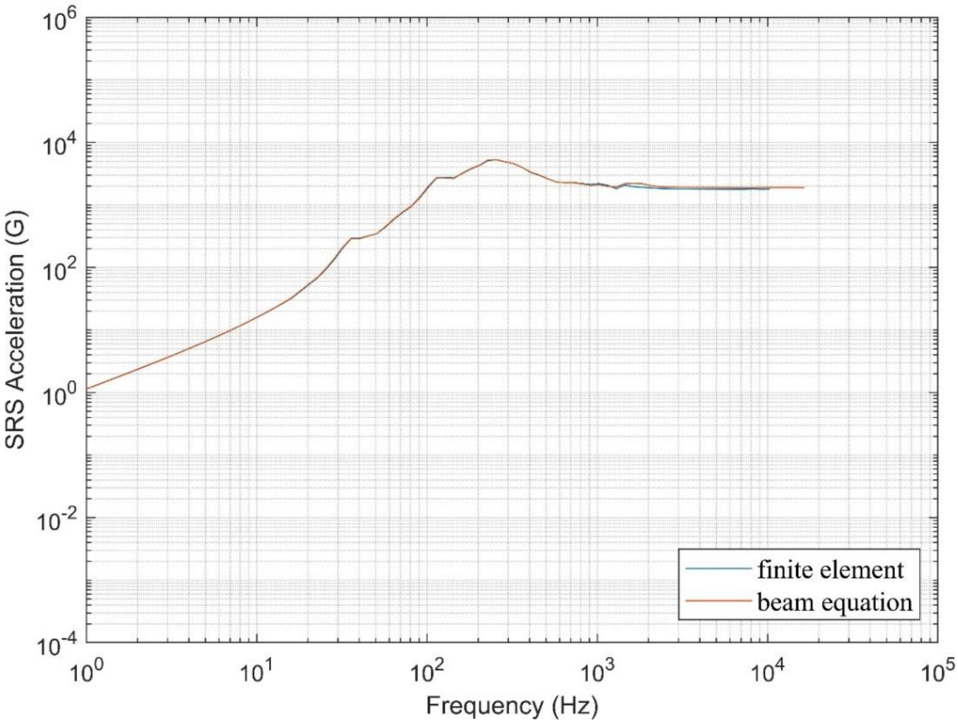


Figure 7.14: Comparison of SRS of fixed-pinned beam No.16 from FEA and beam equation

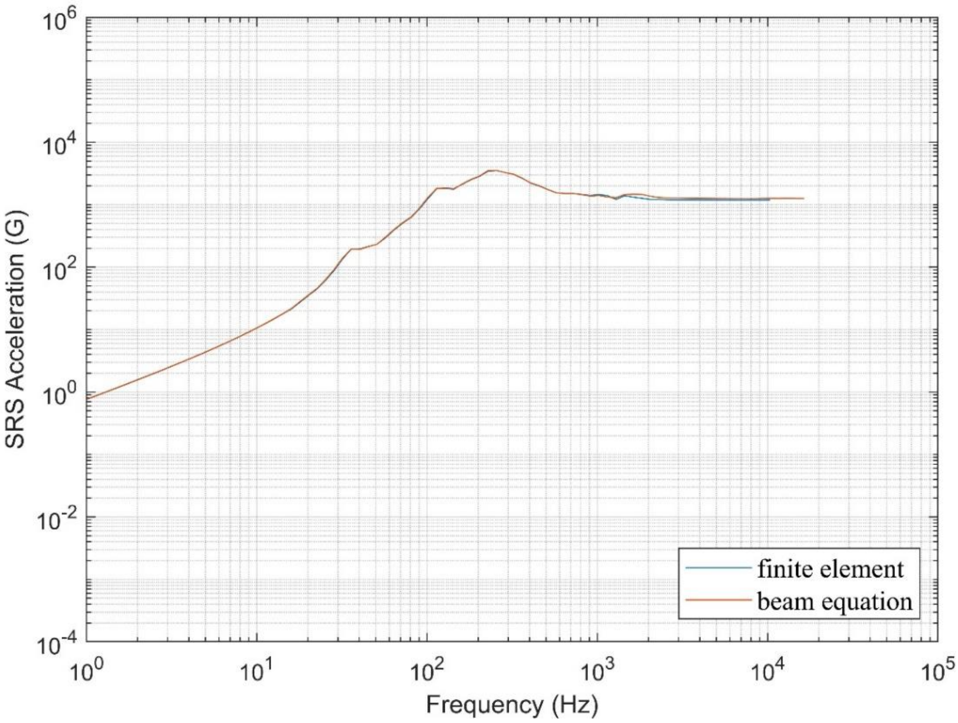


Figure 7.15: Comparison of SRS of fixed-pinned beam No.17 from FEA and beam equation

Appendix 2: Comparison between solid elements and beam elements

Cantilever beams

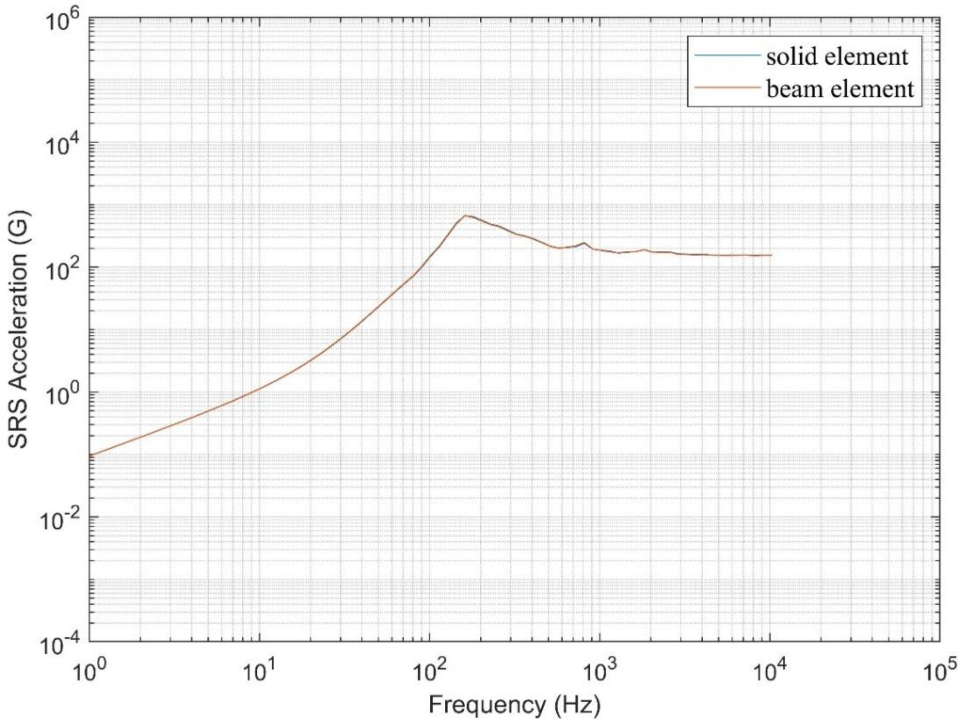


Figure 7.16: Comparison of SRS of cantilever beam No.2 from solid elements and beam elements

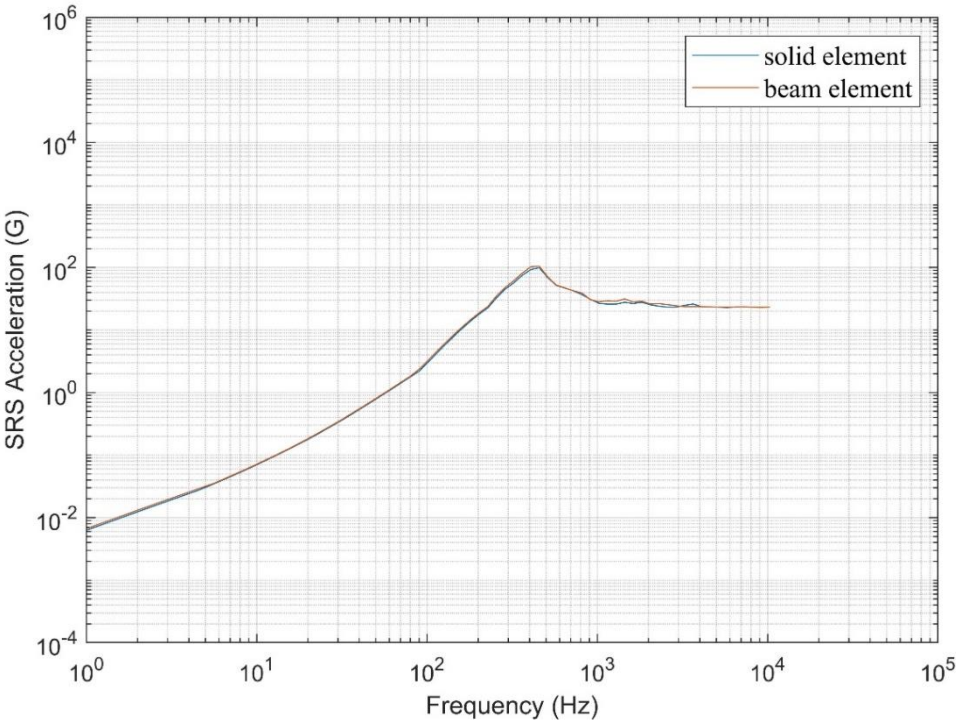


Figure 7.17: Comparison of SRS of cantilever beam No.3 from solid elements and beam elements

Fixed-pinned beams

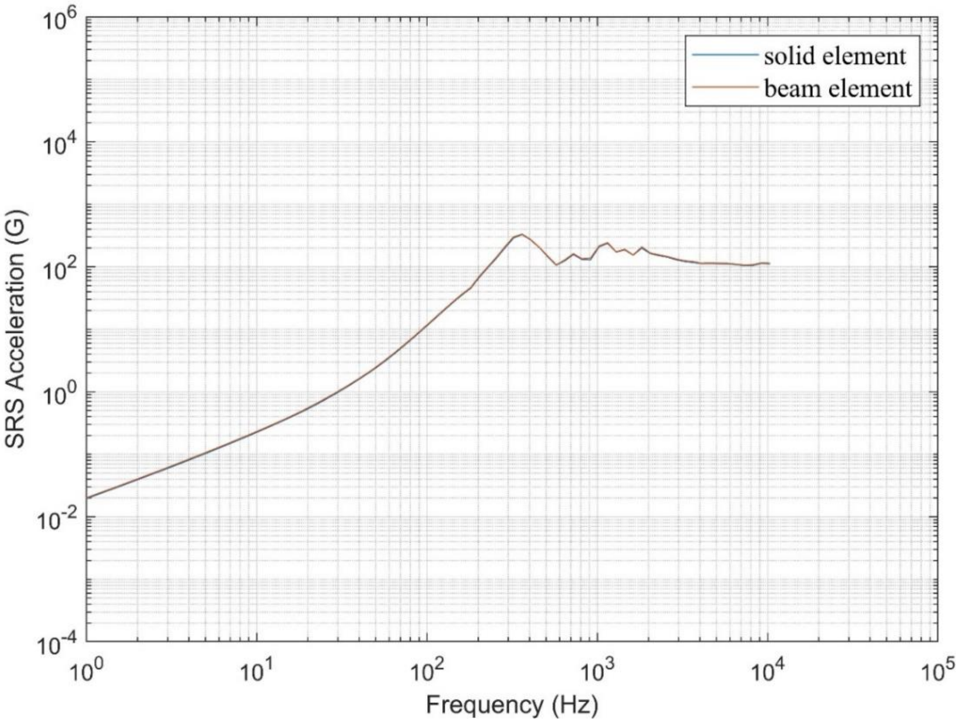


Figure 7.18: Comparison of SRS of fixed-pinned beam No.19 from solid elements and beam elements

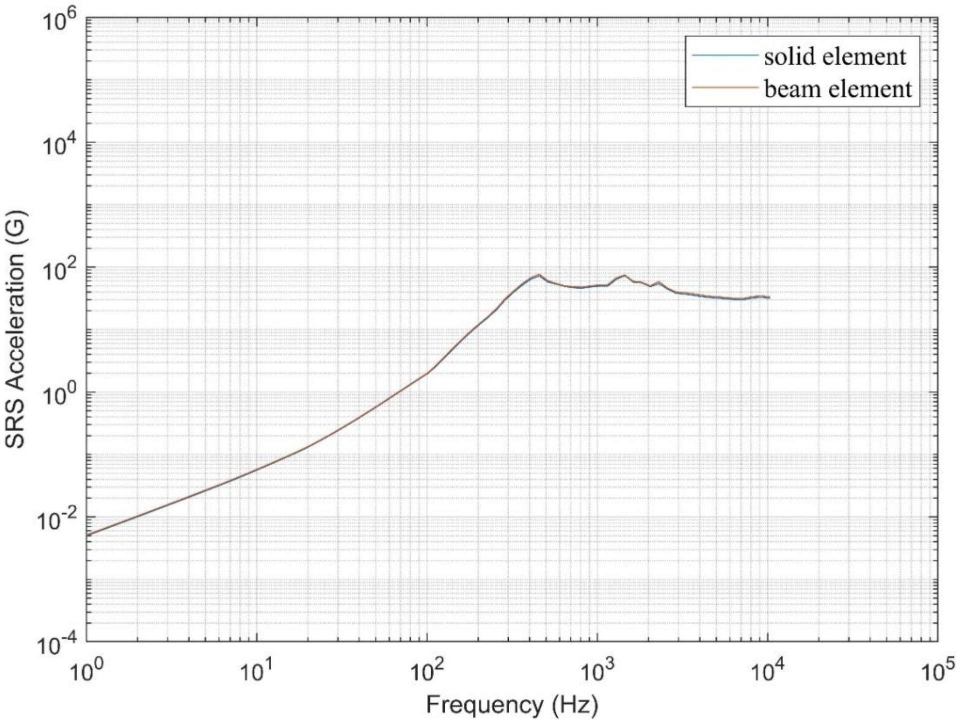


Figure 7.19: Comparison of SRS of fixed-pinned beam No.20 from solid elements and beam elements

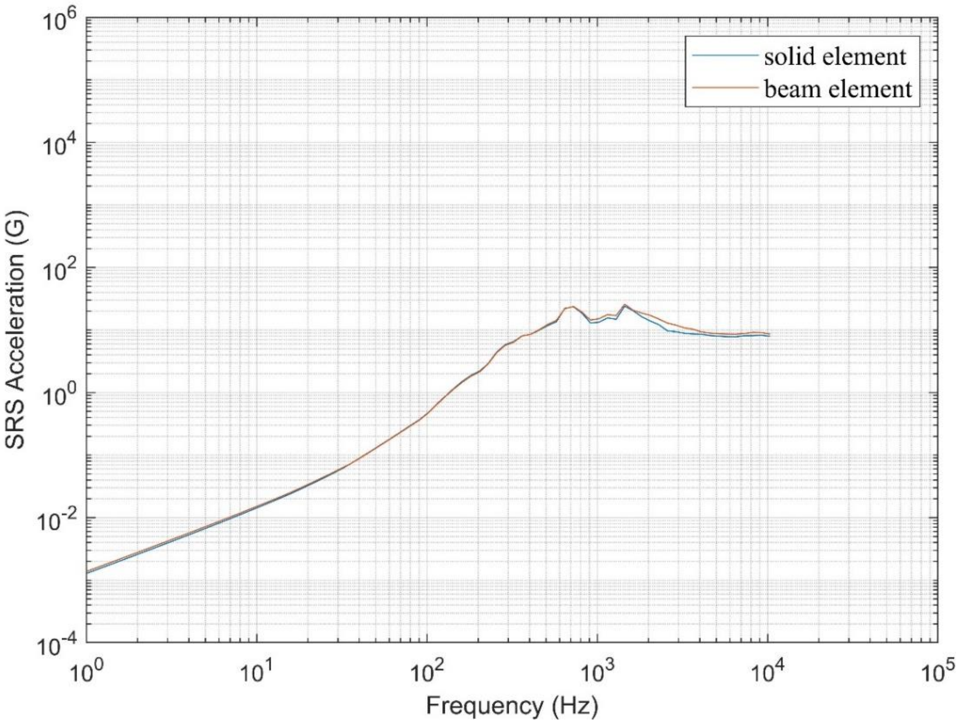


Figure 7.20: Comparison of SRS of fixed-pinned beam No.21 from solid elements and beam elements

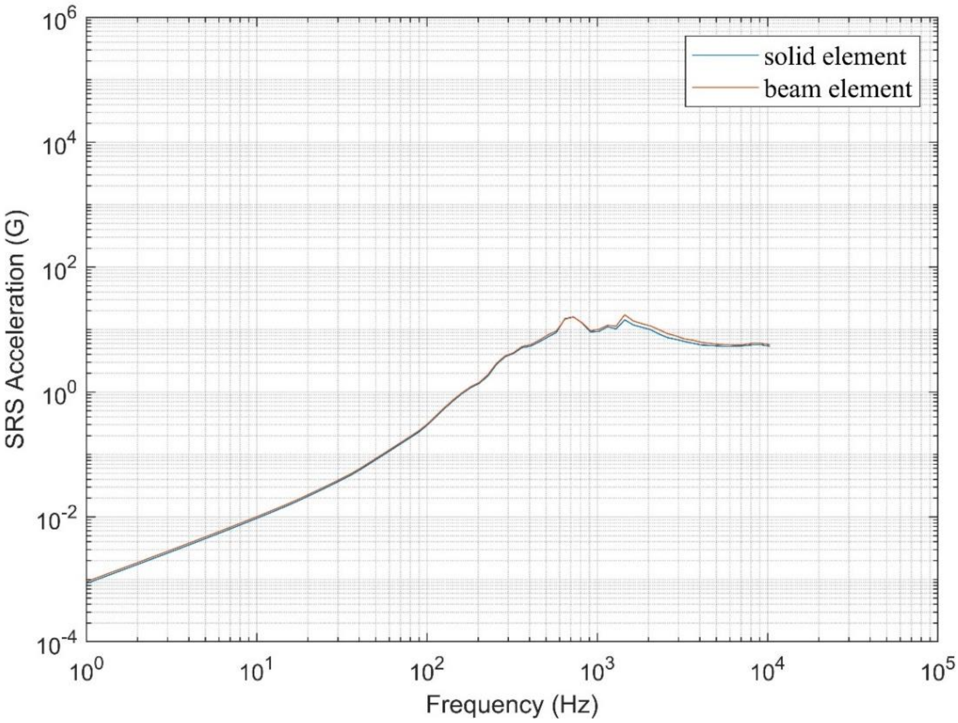


Figure 7.21: Comparison of SRS of fixed-pinned beam No.22 from solid elements and beam elements

TRITA-SCI-GRU 2021:035